

Table 1.4. The compositions (in weight%) of some common wrought superalloys

Alloy	Cr	Co	Mo	W	Nb	Al	Ti	Ta	Fe	Hf	C	B	Zr	Ni
Alloy 10	11.5	15	2.3	5.9	1.7	3.8	3.9	0.75	—	—	0.030	0.020	0.05	Bal
Astrolloy	15.0	17.0	5.3	—	—	4.0	3.5	—	—	—	0.06	0.030	—	Bal
C-263	16	15	3	1.25	—	2.50	5.0	—	—	—	0.025	0.018	—	Bal
Hastelloy S	15.5	—	14.5	—	—	0.3	—	—	1.0	—	—	0.009	—	Bal
Hastelloy X	22.0	1.5	9.0	0.6	—	0.25	—	—	18.5	—	0.10	—	—	Bal
Haynes 230	22.0	—	2.0	14.0	—	0.3	—	—	—	—	0.10	—	—	Bal
Haynes 242	8.0	2.5	25.0	—	—	0.25	—	—	2.0	—	0.15	0.003	—	Bal
Haynes R-41	19.0	11.0	10.0	—	—	1.5	3.1	—	5.0	—	0.09	0.006	—	Bal
Incolloy 800	21.0	—	—	—	—	0.38	0.38	—	45.7	—	0.05	—	—	Bal
Incolloy 801	20.5	—	—	—	—	—	1.13	—	46.3	—	0.05	—	—	Bal
Incolloy 802	21.0	—	—	—	—	0.58	0.75	—	44.8	—	0.35	—	—	Bal
Incolloy 909	—	13.0	—	—	4.7	0.03	1.5	—	42.0	—	0.01	—	—	Bal
Incolloy 925	20.5	—	—	—	—	0.20	2.1	—	29.0	—	0.01	—	—	Bal
Inconel 600	15.5	—	—	—	—	—	—	—	8.0	—	0.08	—	—	Bal
Inconel 601	23.0	—	—	—	—	1.4	—	—	14.1	—	0.05	—	—	Bal
Inconel 617	22.0	12.5	9.0	—	—	1.0	0.3	—	—	—	0.07	—	—	Bal
Inconel 625	21.5	—	9.0	—	3.6	0.2	0.2	—	2.5	—	0.05	—	—	Bal
Inconel 690	29.0	—	—	—	—	—	—	—	9.0	—	0.025	—	—	Bal
Inconel 706	16.0	—	—	—	2.9	0.2	1.8	—	40.0	—	0.03	—	—	Bal
Inconel 718	19.0	—	3.0	—	5.1	0.5	0.9	—	18.5	—	0.04	—	—	Bal
Inconel 738	16.0	8.5	1.75	2.6	0.9	3.4	3.4	1.7	—	—	0.11	0.01	0.05	Bal
Inconel 740	25.0	20.0	0.5	—	2.0	0.9	1.8	—	0.7	—	0.03	—	—	Bal
Inconel X750	15.5	—	—	—	1.0	0.7	2.5	—	7.0	—	0.04	—	—	Bal
LSHR	13	21	2.7	4.3	1.5	3.5	3.5	1.6	—	—	0.030	0.030	0.050	Bal
ME3	13.1	18.2	3.8	1.9	1.4	3.5	3.5	2.7	—	—	0.030	0.030	0.050	Bal
MERL-76	12.4	18.6	3.3	—	1.4	0.2	4.3	—	—	0.35	0.050	0.03	0.06	Bal
Nimonic 75	19.5	—	—	—	—	—	0.4	—	3.0	—	0.10	—	—	Bal
Nimonic 80A	19.5	—	—	—	—	1.4	2.4	—	—	—	0.06	0.003	0.06	Bal
Nimonic 90	19.5	16.5	—	—	—	1.5	2.5	—	—	—	0.07	0.003	0.06	Bal
Nimonic 105	15.0	20.0	5.0	—	—	4.7	1.2	—	—	—	0.13	0.005	0.10	Bal
Nimonic 115	14.3	13.2	—	—	—	4.9	3.7	—	—	—	0.15	0.160	0.04	Bal
Nimonic 263	20.0	20.0	5.9	—	—	0.5	2.1	—	—	—	0.06	0.001	0.02	Bal
Nimonic 901	12.5	—	5.75	—	—	0.35	2.9	—	—	—	0.05	—	—	Bal

Table 1.4. (cont.)

Alloy	Cr	Co	Mo	W	Nb	Al	Ti	Ta	Fe	Hf	C	B	Zr	Ni
Nimonic PE16	16.5	1.0	1.1	—	—	1.2	1.2	—	33.0	—	0.05	0.020	—	Bal
Nimonic PK33	18.5	14.0	7.0	—	—	2.0	2.0	—	0.3	—	0.05	0.030	—	Bal
N18	11.5	15.7	6.5	0.6	—	4.35	4.35	—	—	0.45	0.015	0.015	0.03	Bal
Pyromet 860	13.0	4.0	6.0	—	0.9	1.0	3.0	—	28.9	—	0.05	0.01	—	Bal
Pyromet 31	22.7	—	2.0	—	1.1	1.5	2.5	—	14.5	—	0.04	0.005	—	Bal
Rene 41	19.0	11.0	1.0	—	—	1.5	3.1	—	—	—	0.09	0.005	—	Bal
Rene 88DT	16.0	13.0	4.0	4.0	0.7	2.1	3.7	—	—	—	0.03	0.015	0.03	Bal
Rene 95	14.0	8.0	3.5	3.5	3.5	3.5	2.5	—	—	—	0.15	0.010	0.05	Bal
Rene 104	13.1	18.2	3.8	1.9	1.4	3.5	3.5	2.7	—	—	0.030	0.030	0.050	Bal
RR1000	15.0	18.5	5.0	—	1.1	3.0	3.6	2.0	—	0.5	0.027	0.015	0.06	Bal
Udimet 500	18.0	18.5	4.0	—	—	2.9	2.9	—	—	—	0.08	0.006	0.05	Bal
Udimet 520	19.0	12.0	6.0	1.0	—	2.0	3.0	—	—	—	0.05	0.005	—	Bal
Udimet 630	18.0	—	3.0	3.0	6.5	0.5	1.0	—	18.0	—	0.03	—	—	Bal
Udimet 700	15.0	17.0	5.0	—	—	4.0	3.5	—	—	—	0.06	0.030	—	Bal
Udimet 710	18.0	15.0	3.0	1.5	—	2.5	5.0	—	—	—	0.07	0.020	—	Bal
Udimet 720	17.9	14.7	3.0	1.25	—	2.5	5.0	—	—	—	0.035	0.033	0.03	Bal
Udimet 720LI	16.0	15.0	3.0	1.25	—	2.5	5.0	—	—	—	0.025	0.018	0.05	Bal
Waspaloy	19.5	13.5	4.3	—	—	1.3	3.0	—	—	—	0.08	0.006	—	Bal

elements Al and Ti are generally lower, and Ta is not generally employed. Appreciable quantities of Fe are sometimes employed, and quantities of Nb are present. The Cr content is usually at least 15 wt%, higher than for the cast superalloys. Furthermore, additions such as Re and Ta are not generally made. In Chapters 3 and 4, the compositions of the cast and wrought forms of the superalloys are discussed in detail, and the differences rationalised.

1.3.3 Nickel as a high-temperature material: justification

Although the superalloys display excellent properties at elevated properties, it is not immediately obvious why nickel is a suitable solvent for a high-temperature alloy. Can nickel's role be justified?

Under conditions of high-temperature deformation, one might expect the creep shear strain rate, $\dot{\gamma}$, of a pure metal such as nickel to be proportional to the volume diffusivity, of activation energy Q_v and pre-exponential term $D_{0,v}$. This assumption is considered further in Chapter 2. Thus following ref. [16] one has

$$\dot{\gamma} \propto D_{0,v} \exp \left\{ -\frac{Q_v}{RT} \right\} \quad (1.12)$$

In order to help in the comparison of different material classes, it is helpful to normalise T by the melting temperature, T_m , and $\dot{\gamma}$ by $D_{T_m}/\Omega^{2/3}$, where D_{T_m} is the diffusivity at the melting temperature and Ω is the atomic volume. Hence

$$\bar{\gamma} = \frac{\dot{\gamma}\Omega^{2/3}}{D_{T_m}} \propto \Omega^{2/3} \exp \left\{ -\frac{Q_v}{RT_m} \left(\frac{T_m}{T} - 1 \right) \right\} \quad (1.13)$$

where $\bar{\gamma}$ is the dimensionless shear strain rate. One can see that the most important dimensionless groups are the homologous temperature, T/T_m , and the combination $Q_v/(RT_m)$. Also relevant is the normalising parameter, $D_{T_m}/\Omega^{2/3}$.

What does this analysis reveal? If one assumes for the moment that the combination $Q_v/(RT_m)$ is approximately the same for the different materials under consideration for this application, then for the best high-temperature performance (implying a small value of $\bar{\gamma}$) it is necessary to work at a low homologous temperature T/T_m ; this emphasises the importance of materials which melt at high temperatures. Alternatively, if one accepts the need to work at a particular T/T_m , for example, $T = 0.8 T_m$, then crystal classes which display the *maximum* $Q_v/(RT)$ and *minimum* $D_{T_m}/\Omega^{2/3}$ are the ones which will display the best properties.

We consider first the place of nickel in the family of metallic elements within the periodic table. The melting temperature of the elements shows a strong and remarkable correlation with atomic number (see Figure 1.19). If one focusses attention on the transition metals, then (i) there is a *maximum* in T_m somewhere towards the centre of each row of transition metals, for example, at V, Mo and W for the first, second and third rows, respectively, and (ii) the melting temperatures of the elements in each of the 3d, 4d and 5d rows increases generally as one descends a column from the 3d $\rightarrow$ 4d $\rightarrow$ 5d rows. However, due to the bonding characteristics induced by the increasing number of d electrons, the crystal structure also shows a strong correlation with position within the periodic table. For the transition

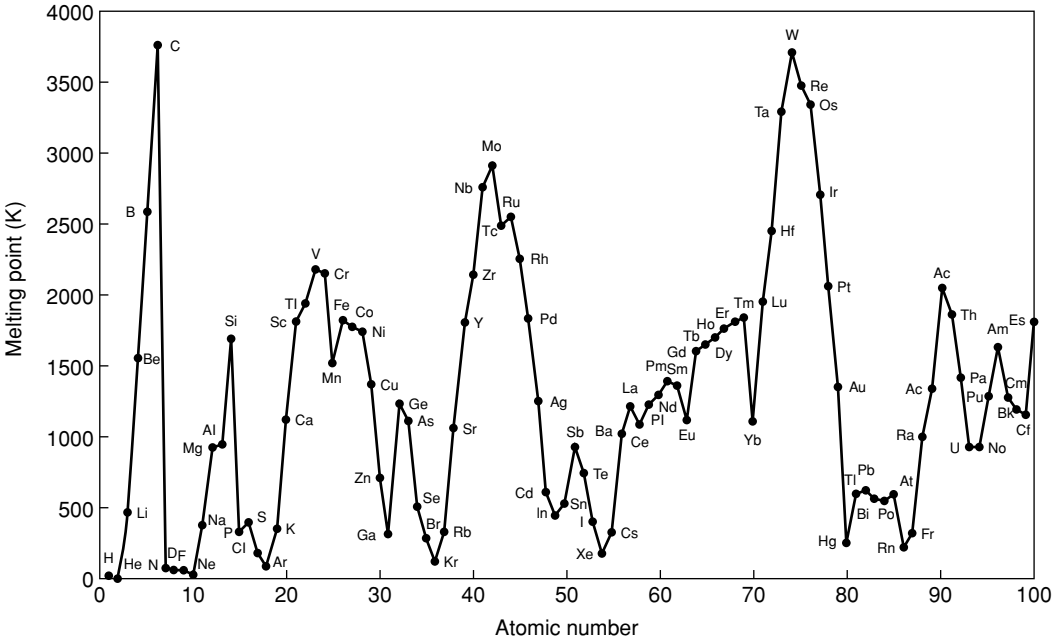


Fig. 1.19. Variation of the melting temperatures of the elements with atomic number.

BCC									
IIIB	IVB	VB	VIB	VIIB	VIII B			IB	IIB
21 Sc 44.956	22 Ti 47.90	23 V 50.942	24 Cr 51.996	25 Mn 54.9380	26 Fe 55.847	27 Co 58.9332	28 Ni 58.71	29 Cu 63.54	30 Zn 65.37
39 Y 88.905	40 Zr 91.22	41 Nb 92.906	42 Mo 95.94	43 Tc [99]	44 Ru 101.07	45 Rh 102.905	46 Pd 106.4	47 Ag 107.870	48 Cd 112.40
* 57 La 138.91	72 Hf 178.49	73 Ta 180.948	74 W 183.85	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.09	79 Au 196.967	80 Hg 200.59
HCP					FCC				
Liquid									

Fig. 1.20. Correlation of the crystal structures of the transition metals with position in the periodic table.

metals (see Figure 1.20), the face-centered cubic (FCC) metals lie towards the far east of the period (groups VIII and 1B), hexagonally close-packed (HCP) metals are at the centre (group VIIB) and the body-centered cubic (BCC) metals are at the far west (groups VB, VIB). Clearly, if one prefers for this application an FCC metal on the grounds that it will be ductile and tough, then the number of available metals is limited; moreover, the list includes

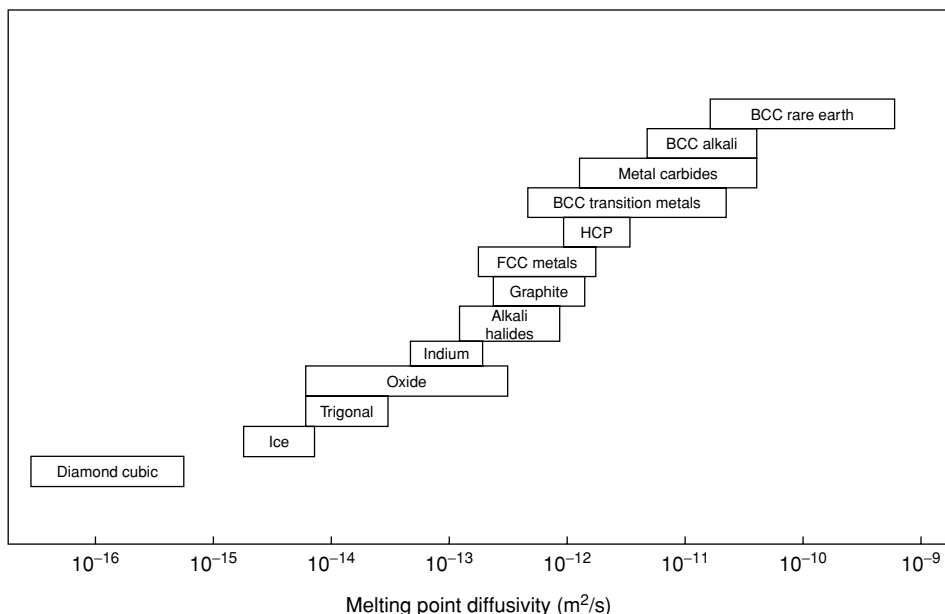


Fig. 1.21. Values of the diffusivities for the various crystal classes at their melting points. Adapted from ref. [17].

many of the platinum group metals (PGMs), which are characterised by their high density and significant cost.

In fact, it turns out that if one widens the search to include materials of different crystal classes, considering oxides and carbide ceramic systems for example, then there is quite a spread in the normalised activation energies, $Q_v/(RT_m)$, and the melting point diffusivity, D_{T_m} . Data for various crystal structures are given in Figures 1.21 and 1.22 for $Q_v/(RT_m)$ and D_{T_m} , respectively [17]. The FCC metals display very high and low values, respectively, and these are considerably better than for the BCC and the HCP metals, at $Q_v/(RT_m) = 18.4$ (compare 17.3 for HCP metals and 17.8 for the BCC metals) and $D_{T_m} = 5 \times 10^{-13} \text{ m}^2/\text{s}$ (compare $\sim 2 \times 10^{-12} \text{ m}^2/\text{s}$ for HCP and $10^{-12} - 10^{-11} \text{ m}^2/\text{s}$ for the BCC transition metals). Other observations support these conclusions. For example, the BCC $\rightarrow$ FCC and BCC $\rightarrow$ HCP transformations in Fe and Ti, respectively, cause a 100-fold reduction in the measured diffusivity.

Given these considerations, one can list the reasons why nickel-based alloys have emerged as the ones chosen for high-temperature applications. First, nickel displays the FCC crystal structure and is thus both *tough* and *ductile*, due to a considerable cohesive energy arising from the bonding provided by the outer d electrons. Furthermore, nickel is stable in the FCC form from room temperature to its melting point, so that there are no phase transformations to cause expansions and contractions which might complicate its use for high-temperature components. Other metals in the transition metal series which display this crystal structure, i.e. the platinum group metals (PGMs), are dense and very expensive. Second, low rates of thermally activated creep require low rates of diffusion – as suggested by the correlation

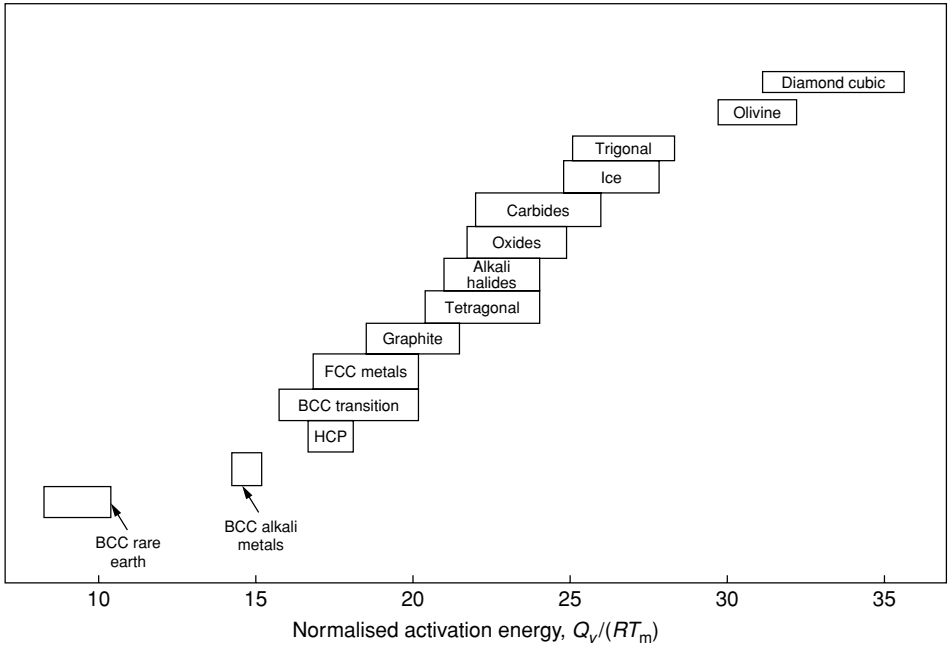


Fig. 1.22. Values of the normalised activation energy for diffusion for the various crystal classes. Adapted from ref. [17].

between the activation energies for self-diffusion and creep in the pure metals. Diffusion rates for FCC metals such as Ni are low; hence, considerable microstructural stability is imparted at elevated temperatures. Finally, consider other elements which possess different crystal structures. Of the HCP metals, only Co displays an acceptable density and cost; Re and Ru are PGMs and are therefore expensive; Os has an oxide which is poisonous; and Tc is radioactive. Co-based superalloys are, in fact, used for high-temperature applications; however, they tend to be more expensive than the nickel-based superalloys. The BCC metals such as Cr are prone to brittleness, and there is a ductile/brittle transition which means that the toughness decreases significantly with decreasing temperature.

1.4 Summary

With nickel as a solvent, but containing in excess of ten different alloying elements, the nickel-based superalloys have emerged as the materials of choice for high-temperature operation when resistance to creep, fatigue and environmental degradation is required. Their metallurgical development is linked inextricably to the history of the jet engine, for which the first superalloy grades were designed, although use has now spread to other high-temperature applications – notably to the turbines used for electricity generation. The technological incentive for further improvements being sought in alloy capability and processing developments arises from the enhanced fuel efficiency and reduced emissions expected of

modern turbine systems; consistent with thermodynamic considerations, higher operating temperatures are then required. The risk of failure by fracture or fatigue necessitates that components fabricated from the superalloys are of the highest integrity, so that inspection and lifetime estimation procedures are critical to safe ongoing operation. A survey of the creep performance of available engineering materials confirms the superiority of the superalloys for high-temperature applications. The use of nickel as the solvent for these materials can be justified on account of its face-centered cubic (FCC) crystal structure, its moderate cost and low rates of thermally activated processes.

Questions

- 1.1 What are the roles played by the fan, compressor, combustor and turbine arrangements in a typical gas turbine engine? How do these affect (i) the pressure and (ii) the average temperature of the gas stream? Explain why your findings justify the use of nickel-based superalloys in the combustor and turbine sections, but not in the compressor regions.
- 1.2 In an aeroengine, the low-pressure (LP) turbine blades are always very much longer than the high-pressure (HP) blading. Why?
- 1.3 Sketch the temperature–time cycle experienced by the first row of turbine blades for one take-off/landing cycle. Assume first that one is considering an aeroengine for a civil aircraft such as the Trent 800 or GE90. Discuss how this mission cycle would differ for (i) the turbine arrangement in a military jet and (ii) that in a land-based industrial gas turbine (IGT). Why do these considerations have implications for the alloy used, particularly the creep and fatigue properties?
- 1.4 A turbojet is operating with a turbine entry temperature (TET) of 1400 K, an overall pressure ratio (OPR) of 25 and an air mass intake, $\dot{M}$, of 20 kg/s. Assuming an ideal cycle such that the turbine work equals the work of compression, determine the *useful work* available to provide a propulsive force. Hence estimate the thermal efficiency, η_{th} . State your assumptions.

Take $c_p = 1000 \text{ J/(kg K)}$, intake temperature = 288 K, $\gamma = 1.4$ and atmospheric pressure = 101 325 Pa. Assume adiabatic isentropic reversible processes such that

$$\frac{T}{T_{ref}} = \left(\frac{P}{P_{ref}} \right)^{(\gamma-1)/\gamma}$$

- 1.5 Using the concepts developed in Question 1.4, investigate how the thermal efficiency, η_{th} , depends upon the TET and the OPR. Which effect is stronger? Begin by making calculations with the OPR set at values of 15, 20 and 25 and the TET at 1200, 1300 and 1400 K. Then make calculations outside of these ranges. What is the relevance of your findings to turbine engine performance? Why, in practice, is there a stronger influence from the TET than predicted by the calculations?
- 1.6 From the definition of the thrust, F_N , as $\dot{M}(V_{jet} - V_0)$ and the propulsive efficiency, η_{prop} , as $F_N V_0 / \frac{1}{2} \dot{M}(V_{jet}^2 - V_0^2)$, where V_0 is the gas entry velocity, show that

alternative expressions are

$$\eta_{\text{prop}} = \frac{2}{1 + V_{\text{jet}}/V_0} = \frac{1}{1 + F_N/(2V_0\dot{M})}$$

- 1.7 Develop ‘performance diagrams’ for the turbojet considered in Questions 1.4–1.6. First plot the thrust, F_N , against the specific fuel consumption, defined as $Z/(F_N C)$. The term C is the calorific value of the fuel (taken as 43 MJ/kg) and Z is the ideal work of combustion defined as $\dot{M} c_p \Delta T$, where ΔT is the difference between the turbine entry and compressor exit temperatures. Take the entry velocity, V_0 , as 250 m/s. On your diagram, plot contours corresponding to specific values of the TET and OPR.

As a second exercise, plot the overall efficiency, $\eta_{\text{overall}} = \eta_{\text{th}} \eta_{\text{prop}}$, versus OPR for various values of the TET. Comment on your findings.

- 1.8 One factor in setting the fan diameter of a modern turbofan such as the Trent 800 or GE90 is the clearance of the engine beneath the wing of the aircraft – this needs to be sufficient for take-off. Use these considerations to estimate (i) an approximate value of the fan diameter, (ii) the length of the fan blades, if the ratio of inner and outer diameter is constrained to be 0.37, (iii) the rate of mass intake into the engine, given your dimensions and a typical cruising speed and (iv) the mass intake (in kg/s) into the turbine section of the engine, if the bypass ratio (BPR) is 6:1.

The shaft power developed by a turbine stage is given approximately by $\dot{M} c_p \Delta T$, where $\dot{M}$ is the mass flow (kg/s), c_p is the heat capacity of 1000 J/(kg K) and ΔT is the temperature drop across the blading. By estimating the temperature drop to be expected in the turbine, compare the power developed by the high-pressure (HP) and low-pressure (LP) stages. Take the turbine entry temperature (TET) to be 1700 K. If the HP and LP shafts spin at 1000 and 300 rad/s, respectively, estimate the power developed in the two shafts, and the associated torques. [*Hint.* The shaft power is torque multiplied by angular speed.]

- 1.9 The constant, C , in the Larson–Miller parameter, P , which is defined according to $P = T\{C + \log_{10} t_r\}$, is often taken to be 20. Give some justification for this.
- 1.10 The costs of superalloy components are often considered to be an important consideration. Estimate the relative costs of (a) a turbine blade cast in single-crystal, columnar and equiaxed forms and (b) a turbine disc fabricated from powder metallurgy and conventional cast-and-wrought processing.
- 1.11 The best choice of metal solvent for a high-temperature alloy is often considered to be that with the highest value of $Q_v/(RT_m)$, where Q_v is the activation energy for lattice diffusion, R is the gas constant and T_m is the melting temperature. On this basis, FCC metals (with a value of 18.4) perform better than BCC metals (17.8) which in turn are superior to HCP ones (17.3). An alternative measure is the diffusivity at the melting temperature.

These suggestions, however, depend upon a normalisation against the melting temperature, T_m , whereas, for gas turbine applications, one could argue that (i) the absolute metal temperature is more relevant and (ii) the density and cost should be

accounted for. Re-examine the properties of the transition metals with these considerations in mind. Suggest a merit criterion for turbine blade applications, and find the transition metal which maximises it. [Hint. Make use of the considerations of Brown and Ashby (see ref. [17]). For FCC metals, the pre-exponential factor is 5.4×10^{-5} ; for BCC it is 1.6×10^{-4} , and for HCP it is 4.9×10^{-5} . All units are in m^2/s .]

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2 The physical metallurgy of nickel and its alloys

Nickel is the fifth most abundant element on earth. The atomic number is 28, and this places it in the first row of the d block of transition metals, alongside iron and cobalt. The atomic weight is 58.71, the weighted average of the five stable isotopes 58, 60, 61, 62 and 64, which are found with probabilities 67.7%, 26.2%, 1.25%, 3.66% and 1.16%, respectively. The crystal structure is face-centred cubic (FCC; see Figure 2.1), from ambient conditions to the melting point, 1455 °C, which represents an absolute limit for the temperature capability of the nickel-based superalloys. The density under ambient conditions is 8907 kg/m³. Thus, compared with other metals used for aerospace applications, for example, Ti (4508 kg/m³) and Al (2698 kg/m³), Ni is rather dense. This is a consequence of a small interatomic distance, arising from the strong cohesion provided by the outer d electrons – a characteristic of the transition metals.

In this chapter, some important aspects of the physical metallurgy of nickel and its alloys are considered. Section 2.1 is concerned with the compositions of the superalloys and the phases promoted by the presence of the alloying elements. Such composition–microstructure relationships have been established over many years, and considerable use of them is required when designing new grades of superalloy. Much of this quantitative information now resides in thermodynamic databases, which are capable of predicting the equilibrium fractions and compositions of the metallurgical phases as a function of mean chemistry and temperature; the background to these methods is presented. One of these – the ordered γ' phase – plays a pivotal role in superalloy metallurgy, and its characteristics are introduced here. In Section 2.2, attention is focussed on the different lattice defects which arise – it turns out that control of these is vital if significant high-temperature strength and creep resistance are to be achieved. The role of vacancies is emphasised since these play an important role in the mechanism of diffusion, together with the planar defects such as anti-phase boundaries and stacking faults which are involved in the dislocation dissociation reactions which determine the high-temperature properties. Section 2.3 is concerned with the role of the γ' phase in promoting high-temperature strength. First, theory for order strengthening by distributions of the γ' phase is presented. Next, the anomalous yielding effect – the origin of which has proved a controversial topic in superalloy metallurgy – is reviewed. Consideration is given to the role played by the γ' phase in promoting the high-temperature creep strength in these systems. Finally, in Section 2.4, attention is turned to the creep behaviour of nickel and its alloys.

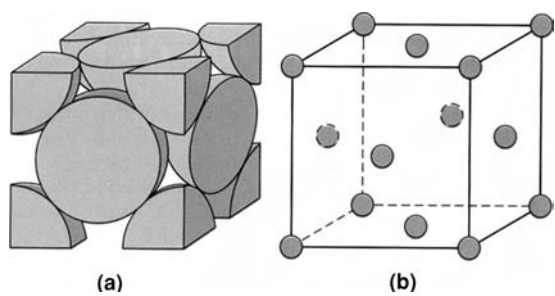


Fig. 2.1. The unit cell of the face-centred cubic (FCC) crystal structure, which is displayed by nickel.

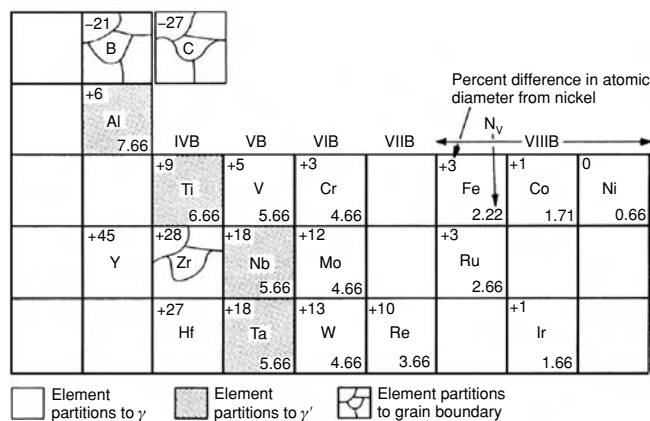


Fig. 2.2. Categories of elements important to the constitution of the nickel-based superalloys, and their relative positions in the periodic table. Adapted from ref. [1].

2.1 Composition–microstructure relationships in nickel alloys

The compositions of some important nickel-based superalloys have been given in Tables 1.3 and 1.4. One can see that the number of alloying elements is often greater than ten and consequently, if judged in this way, the superalloys are amongst the most complex of materials engineered by man. Although a wide variety of alloying elements are used, some broad rules are apparent. Most of the alloys contain significant amounts of chromium, cobalt, aluminium and titanium. Small amounts of boron, zirconium and carbon are often included. Other elements that are added, but not to all alloys, include rhenium, tungsten, tantalum and hafnium, from the 5d block of transition metals, and ruthenium, molybdenum, niobium and zirconium from the 4d block. Certain superalloys, such as IN718 and IN706, contain significant proportions of iron, and should be referred to as nickel–iron superalloys.

Thus, most of the alloying elements are taken from the d block of transition metals [1]. Unsurprisingly, the behaviour of each alloying element and its influence on the phase stability depends strongly upon its position within the periodic table (see Figure 2.2). A

first class of elements includes nickel, cobalt, iron, chromium, ruthenium, molybdenum, rhenium and tungsten; these prefer to partition to the austenitic γ and thereby stabilise it. These elements have atomic radii not very different from that of nickel. A second group of elements, aluminium, titanium, niobium and tantalum, have greater atomic radii and these promote the formation of ordered phases such as the compound $\text{Ni}_3(\text{Al}, \text{Ta}, \text{Ti})$, known as γ' . Boron, carbon and zirconium constitute a third class that tend to segregate to the grain boundaries of the γ phase, on account of their atomic sizes, which are very different from that of nickel. Carbide and boride phases can also be promoted. Chromium, molybdenum, tungsten, niobium, tantalum and titanium are particularly strong carbide formers; chromium and molybdenum promote the formation of borides.

The microstructure of a typical superalloy consists therefore of different phases, drawn from the following list [2].

- (i) The gamma phase, denoted γ . This exhibits the FCC structure, and in nearly all cases it forms a continuous, matrix phase in which the other phases reside. It contains significant concentrations of elements such as cobalt, chromium, molybdenum, ruthenium and rhenium, where these are present, since these prefer to reside in this phase.
- (ii) The gamma prime precipitate, denoted γ' . This forms as a precipitate phase, which is often coherent with the γ -matrix, and rich in elements such as aluminium, titanium and tantalum. In nickel–iron superalloys and those rich in niobium, a related ordered phase, γ'' , is preferred instead of γ' .
- (iii) Carbides and borides. Carbon, often present at concentrations up to 0.2 wt%, combines with reactive elements such as titanium, tantalum and hafnium to form MC carbides. During processing or service, these can decompose to other species, such as M_{23}C_6 and M_6C , which prefer to reside on the γ -grain boundaries, and which are rich in chromium, molybdenum and tungsten. Boron can combine with elements such as chromium or molybdenum to form borides which reside on the γ -grain boundaries.

Other phases can be found in certain superalloys, particularly in the service-aged condition, for example, the topologically close-packed (TCP) phases μ , σ , Laves, etc. However, the compositions of the superalloys are chosen to avoid, rather than to promote, the formation of these compounds.

2.1.1 The FCC phase

Since the FCC phase constitutes the matrix phase for the superalloys, it is desirable to quantify its stability with respect to other possible crystal structures. This is important since any phase transformations, either during thermal cycling or during extended periods of operation, will confer poor high-temperature properties. It is also relevant to the coupling of phase diagram information with thermochemical data, for the analysis of the microstructural stability of the superalloys. One can begin by estimating the driving force required for transformation of pure Ni to a crystal structure other than the FCC form. Consider the binary Ni–Cr binary phase diagram [3] (see Figure 2.3). The solute Cr displays the body-centered cubic (BCC) crystal structure, and extrapolation of the (liquid + BCC) two-phase field on the Cr-rich side of the binary gives a theoretical metastable liquid $\rightarrow$ BCC

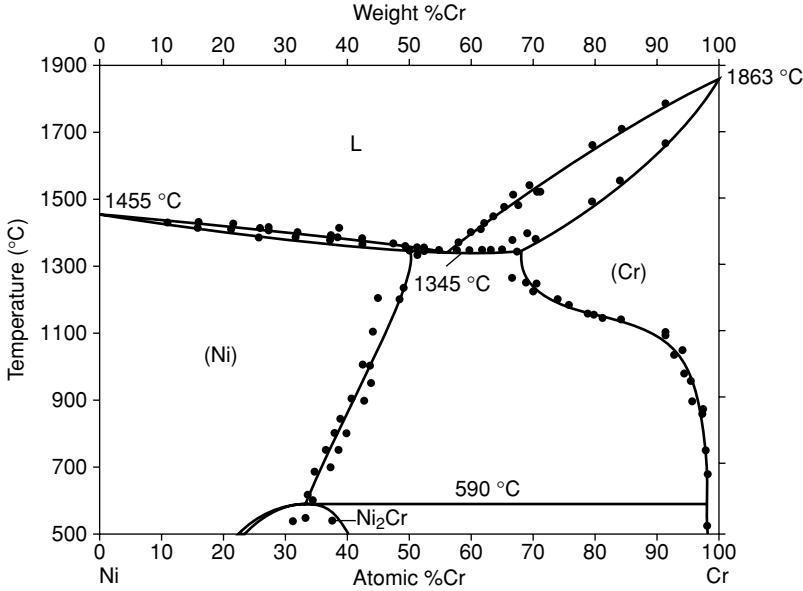


Fig. 2.3. The Ni–Cr binary phase diagram [3]. Note that extrapolation of the liquid + BCC two-phase region to pure Ni gives a metastable liquid to BCC transformation temperature of $\sim 700^\circ\text{C}$.

transformation temperature of $\sim 700^\circ\text{C}$, 750°C below Ni's true melting temperature of 1455°C . For most solidification reactions, the molar entropy of fusion has a magnitude of about $10\text{ J}/(\text{mol K})$, as noted by Trouton's rule, which states that the absolute melting temperature is proportional to the molar enthalpy of fusion. The latent heat of melting is thus about $(1455 + 273) \times 10 \times 10^{-3} = 17.3\text{ kJ/mol}$. It follows that the lattice stability at 750°C of the FCC phase with respect to BCC is approximately given by

$$\Delta G_{\text{Ni}}^{\text{BCC} \rightarrow \text{FCC}} = G_{\text{Ni}}^{\text{FCC}} - G_{\text{Ni}}^{\text{BCC}} = 750 \times (-10) = -7500\text{ J/mol} \quad (2.1a)$$

and that this is a rough estimate of the value at the melting temperature of pure nickel, 1455°C , if Trouton's rule applies. The lattice stabilities of the FCC and BCC phases with respect to the liquid (L) form are thus given by

$$\Delta G_{\text{Ni}}^{\text{L} \rightarrow \text{FCC}} = G_{\text{Ni}}^{\text{FCC}} - G_{\text{Ni}}^{\text{L}} = -17300 + 10 T \text{ J/mol} \quad (2.1b)$$

and

$$\Delta G_{\text{Ni}}^{\text{L} \rightarrow \text{BCC}} = G_{\text{Ni}}^{\text{BCC}} - G_{\text{Ni}}^{\text{L}} = -9800 + 10 T \text{ J/mol} \quad (2.1c)$$

The lattice stability of the hexagonally close-packed (HCP) form of Ni can be estimated in a similar way, for example, by considering the Ni–Ru binary diagram. The lattice stabilities deduced in this simple way are in reasonable agreement with the values suggested in the literature [4] (see Figure 2.4). The values confirm the considerable solid-state stability of the FCC form.

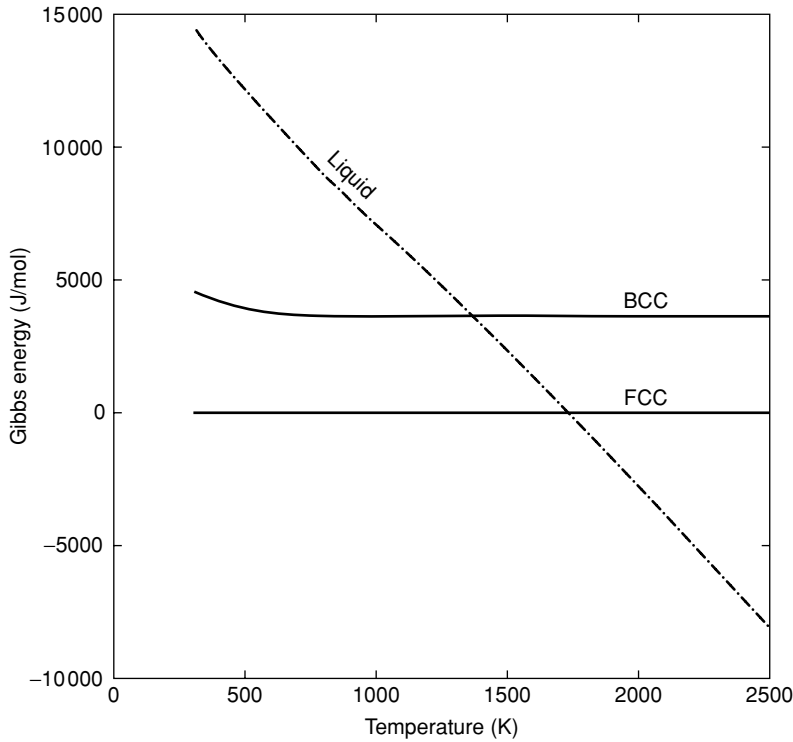


Fig. 2.4. Variation of the liquid and BCC lattice stabilities of Ni, with respect to the FCC form. Data taken from ref. [4].

It is possible to take these calculations further, to justify the stability of the FCC structure of Ni in the context of the crystal structures displayed by the transition metals; see Figure 1.20. To a zeroth-order approximation, the liquid, FCC, BCC and HCP phases can be modelled as ideal [5]. Defining $\Delta G_i^{L \rightarrow \phi} = G_i^\phi - G_i^L$ as the driving force for transformation for the pure constituent i , the two-phase (liquid + ϕ) region bounded by the liquidus and solidus is described by

$$\mu_i^L = G_i^L + RT \ln x_i^L = \mu_i^\phi = G_i^\phi + RT \ln x_i^\phi \quad (2.2a)$$

or, after some manipulation,

$$\frac{x_i^\phi}{x_i^L} = \exp \left\{ -\frac{\Delta G_i^{L \rightarrow \phi}}{RT} \right\} \quad i = \text{solute, solvent} \quad (2.2b)$$

consistent with the equality of chemical potential, μ_i^ϕ , required by equilibrium conditions. The term x_i^ϕ represents the mole fraction of component i in phase ϕ . Thus it is seen that, for the ideal case, each two-phase region on the binary phase diagram requires just four parameters for its description, two for each of the pure constituents: a transformation temperature T_m for the liquid to solid phase transformation (at which the free energy, $\Delta G_i^{L \rightarrow \phi}$, disappears) and an associated entropy of fusion, $\Delta S_i^{L \rightarrow \phi}$. Optimisation of the four

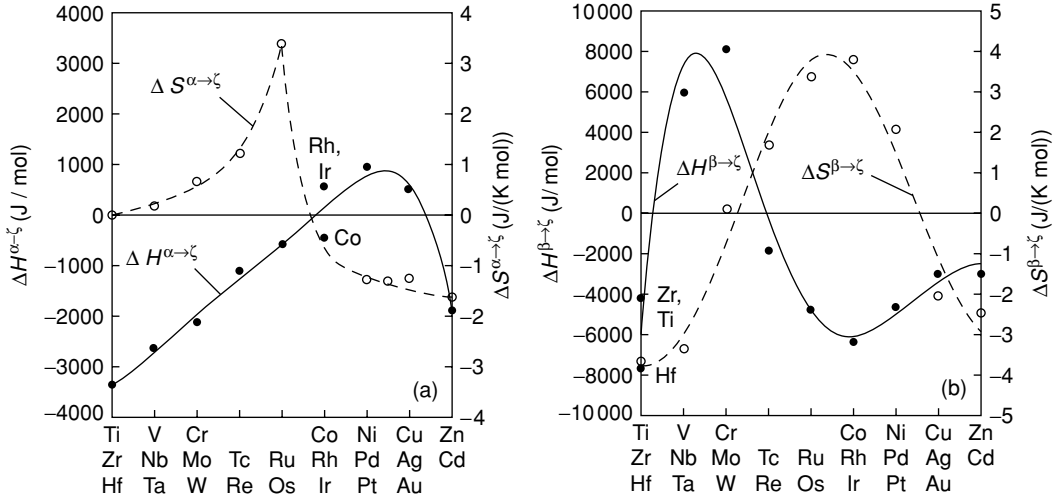


Fig. 2.5. Estimates of the enthalpies and entropies of transformation necessary to convert one mole of transition metal from the (a) FCC to HCP and (b) BCC to HCP forms [5]. The symbols α , ζ and β refer to the FCC, HCP and BCC phases, respectively. In (a), the FCC form is preferred for positive and negative values of the enthalpy and entropy changes, respectively; in (b), the BCC form is preferred for positive and negative values of the enthalpy and entropy changes, respectively. Estimates for the magnetic elements Mn and Fe cannot be made with this simplistic method.

parameters to give a best fit to the available phase diagram information can then be carried out, enabling estimates of $\Delta H_i^{\text{FCC} \rightarrow \text{HCP}}$, $\Delta S_i^{\text{FCC} \rightarrow \text{HCP}}$, $\Delta H_i^{\text{BCC} \rightarrow \text{HCP}}$ and $\Delta S_i^{\text{BCC} \rightarrow \text{HCP}}$ to be made for each transition metal i . It should be noted, however, that this is a lengthy exercise. The values derived in this way [5] are given in Figure 2.5. Predicted correctly is the stability of the BCC, HCP and FCC crystal structures as one proceeds from the far west to the far east of the d block of transition metals.

In practice, the solutions formed by Ni with the other elements are rarely ideal, so that it is seldom accurate to approximate the activity coefficients as unity – thus the above considerations will necessarily contain systematic errors. For example, the liquid phase in the binary Ni–Cr system displays a finite enthalpy of mixing of 2300 J/mol at $x_{\text{Cr}} = 0.5$ at 1873 K [3], when the standard states of both Ni and Cr correspond to the liquid phase. The activities of Cr show negative deviations from ideality up to about $x_{\text{Cr}} = 0.33$, and positive deviations thereafter [3]. This experimental information cannot be reproduced by the ideal solution model. Hence the greater complexity of the regular solution model is more appropriate, in which one includes a molar excess energy, $\Delta G_m^{\text{excess}}$, which is formally identified with the enthalpy of mixing, ΔH_m^{mix} , and an interaction parameter, Ω , such that

$$\Delta G_m^{\text{excess}} = \Delta H_m^{\text{mix}} = \Omega x_i(1 - x_i) \quad (2.3a)$$

The corresponding partial molar quantities can be shown to be

$$\Delta \bar{H}_i = \Delta \bar{G}_i^{\text{excess}} = \Omega(1 - x_i)^2 \quad i = 2 \text{ (solute), and } 1 \text{ (solvent)} \quad (2.3b)$$

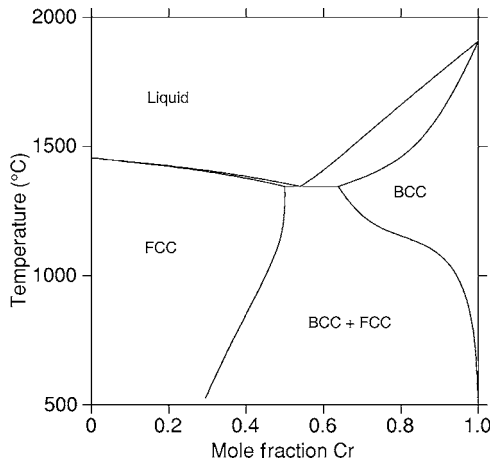


Fig. 2.6. Ni–Cr binary phase diagram calculated using the Thermo-Calc software and a database of thermodynamic parameters.

These must be added to the chemical potentials of Equation (2.2a), so that

$$\mu_i^L = G_i^L + RT \ln x_i^L + \Delta \bar{G}_i^{\text{excess}, L} = \mu_i^\phi = G_i^\phi + RT \ln x_i^\phi + \Delta \bar{G}_i^{\text{excess}, \phi} \quad (2.3c)$$

After some manipulation, one obtains

$$\frac{\Delta G_1^{L \rightarrow \phi}}{RT} + \frac{\Omega^\phi}{RT} (x_2^\phi)^2 - \frac{\Omega^L}{RT} (x_2^L)^2 + \ln \left\{ \frac{(1 - x_2^\phi)}{(1 - x_2^L)} \right\} = 0 \quad (2.4a)$$

and

$$\frac{\Delta G_2^{L \rightarrow \phi}}{RT} + \frac{\Omega^\phi}{RT} (1 - x_2^\phi)^2 - \frac{\Omega^L}{RT} (1 - x_2^L)^2 + \ln \left\{ \frac{x_2^\phi}{x_2^L} \right\} = 0 \quad (2.4b)$$

Equations (2.4a) and (2.4b) represent a set of coupled equations which can be solved, iteratively, for the unknowns x_2^ϕ and x_2^L . The values of Ω^L and Ω^ϕ are chosen such that the available experimental information is reproduced accurately.

For the calculation of the phase diagrams of multicomponent nickel-based superalloys and the associated phase equilibria, there is now available thermodynamic software such as the Thermo-Calc package [6], which enables calculations to be performed quickly and with precision. Such software requires a thermodynamic database of parameters [7] which describe the lattice stabilities and the interaction parameters. These need to be chosen by careful assessment of experimental phase diagram information, using methods of the type described here but also others of greater sophistication. Figures 2.6 and 2.7 illustrate computed Ni–Cr and Ni–Al binary diagrams determined with these so-called CALPHAD (CALculation of PHase Diagrams) methods. Note the very good agreement between the computed Ni–Cr diagram and the experimental one given in Figure 2.3.

The liquidus and solidus lines for the Ni–Re, Ni–W, Ni–Ru, Ni–Co, Ni–Al, Ni–Mo, Ni–Ta and Ni–Ti binary diagrams are plotted in Figure 2.8. One can see that the variation in the liquidus induced by each alloying element i does not correlate with its melting temperature. For example, W and Ta (which lie next to each other in the periodic table)

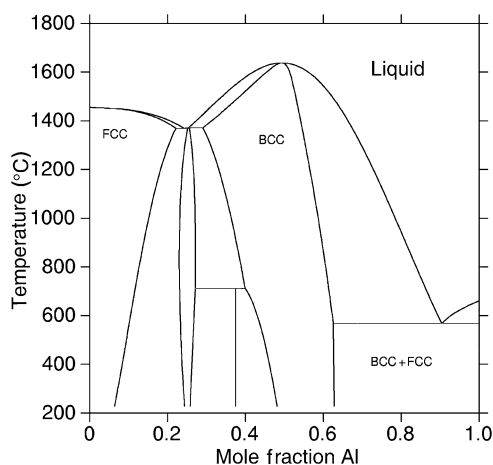


Fig. 2.7. Ni–Al binary phase diagram calculated using the Thermo-Calc software and a database of thermodynamic parameters.

increase and decrease the liquidus temperature by about +5 K/at% and –5 K/at% of solute, respectively. The element Re, however, is notable for its strong positive influence on the liquidus temperature. Consistent with the Hume–Rothery rule, the solubility of i in FCC nickel is found to depend strongly upon element i 's atomic diameter; see Figure 2.9. The solubility is large only when atomic diameter is comparable to that of nickel, as for the platinum group metals Pt, Ir, Pd and Rh.

2.1.2 The gamma prime phase

Although Ni and Al possess the same crystal structure so that complete mutual solid solubility might be anticipated, the binary Ni–Al system exhibits a number of solid phases other than the FCC one, as the phase diagram confirms. These possess the following characteristics: (i) a significant degree of directional, covalent bonding such that precise stoichiometric relationships exist between the number of Ni and Al atoms in each unit cell, and (ii) crystal structures in which Ni–Al rather than Ni–Ni or Al–Al bonds are preferred. Thus a strong degree of chemical order is displayed, and consequently these phases are therefore referred to as *ordered* to distinguish them from the *disordered* solutions based on the FCC or BCC crystal structures. The chemical formulae are Ni_3Al , NiAl , Ni_2Al_3 , NiAl_3 and Ni_2Al_9 . For each compound, the enthalpy of ordering is significant. This is demonstrated by Figure 2.10, in which the enthalpies of formation at 25 °C are plotted against the enthalpy of mixing of the FCC phase [8]. The enthalpy of formation is largest for the β -NiAl compound, which displays the CsCl crystal structure. However, of great significance is the gamma prime (γ') phase, Ni_3Al , particularly in view of the role it plays in conferring strength to the superalloys. It displays the primitive cubic, $L1_2$, crystal structure (see Figure 2.11), with Al atoms at the cube corners and Ni atoms at the centres of the faces. It is notable that each Ni atom has four Al and eight Ni as nearest neighbours, but that each Al atom is co-ordinated by twelve Ni atoms – thus Ni and Al have distinct site occupancies. The data in Figure 2.10 indicate that the ordering energy with respect to the FCC phase is about 3 kJ/mol.

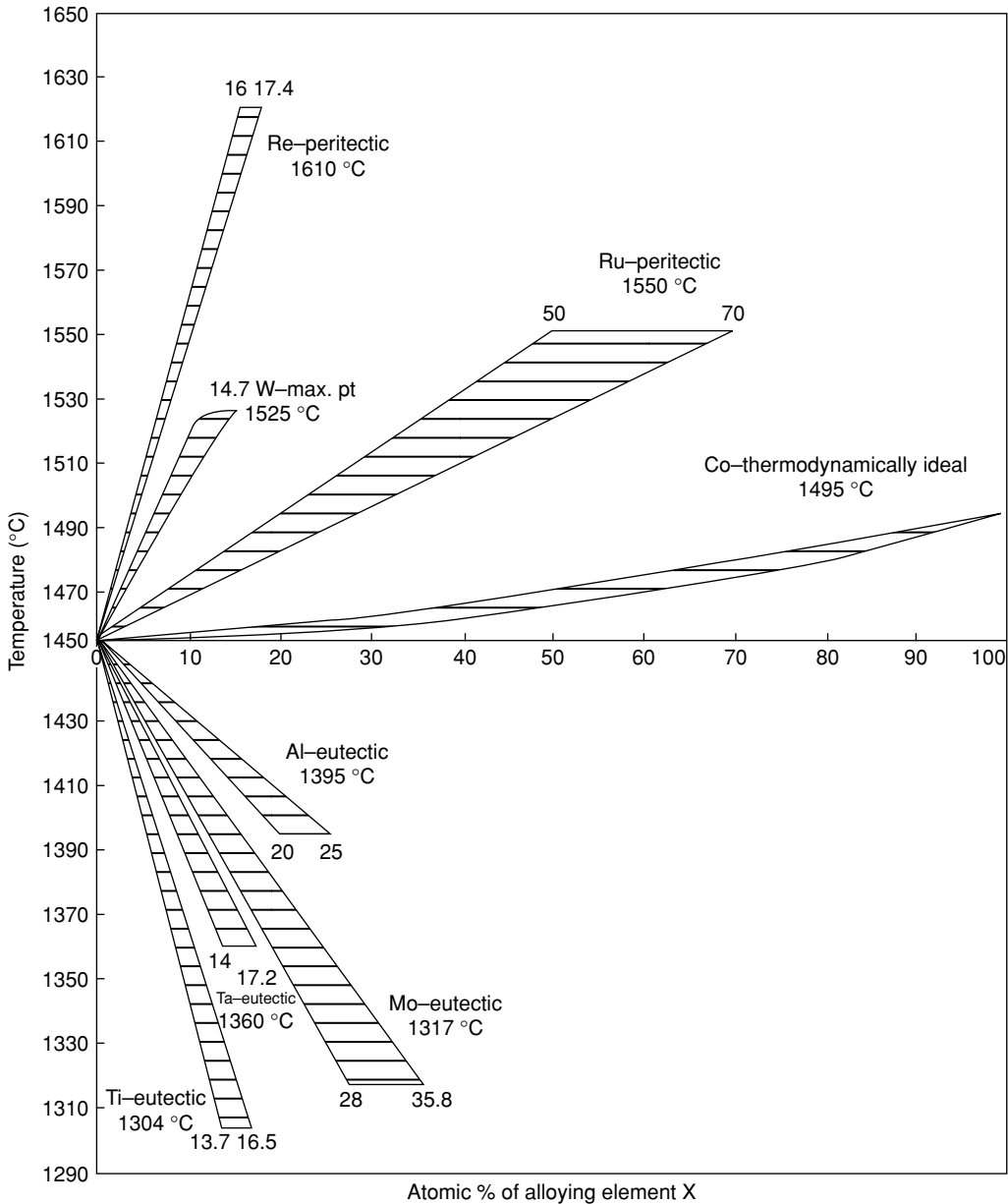


Fig. 2.8. The solidus and liquidus lines in the Ni-X binary systems.

It is interesting to examine the various pieces of evidence which confirm the ordered nature of the γ' phase. Consider first the ternary phase diagrams Ni-Al-X, where X = Co, Cr, Mo, W, etc. In each case, the γ' phase field is extended in a direction which depends upon the solubility of X in the γ' phase [9] (see Figure 2.12). Elements such as Co and Pt promote γ' phase fields which are parallel to the Ni-X axis on the ternary section, implying

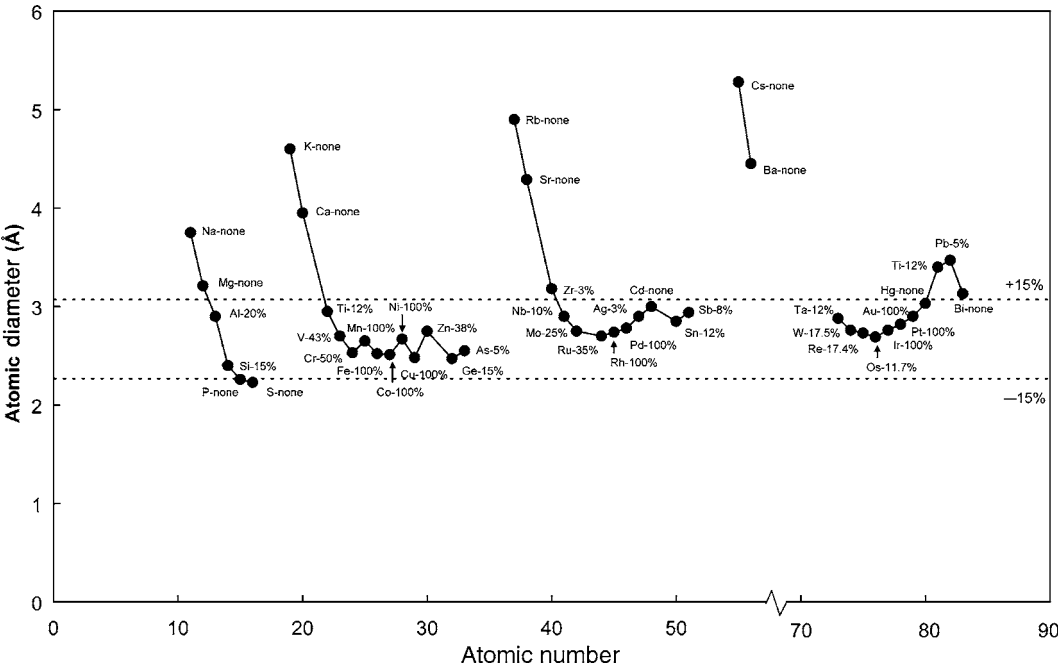


Fig. 2.9. Variation of the atomic diameter of the elements as a function of atomic number; also given are data for the maximum solubility of each element X in FCC nickel. The horizontal lines correspond to a $\pm 15\%$ deviation away from the atomic diameter of nickel. Note that significant solubility occurs only when the atomic sizes of nickel and solute are comparable.

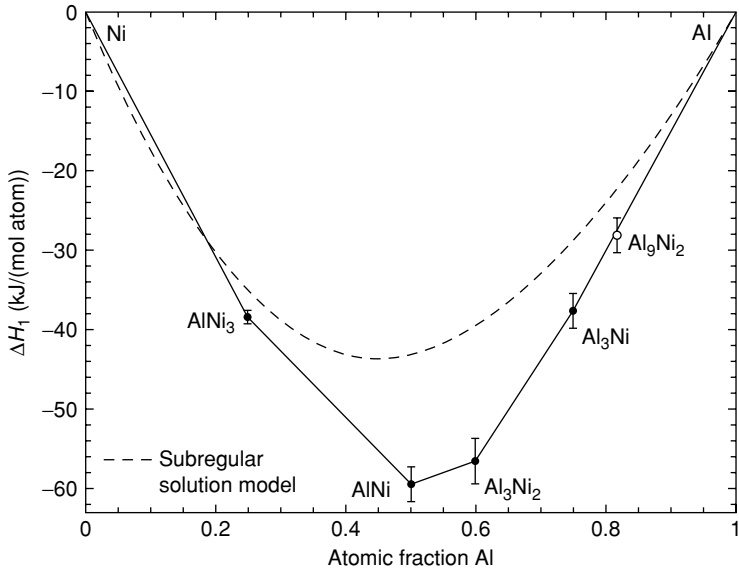


Fig. 2.10. Variation of the enthalpy of formation, ΔH_f , for the various intermetallic compounds in the binary Ni–Al system [8]. The broken line corresponds to the enthalpy of mixing of the disordered FCC phase, with respect to the pure Ni and Al constituents.

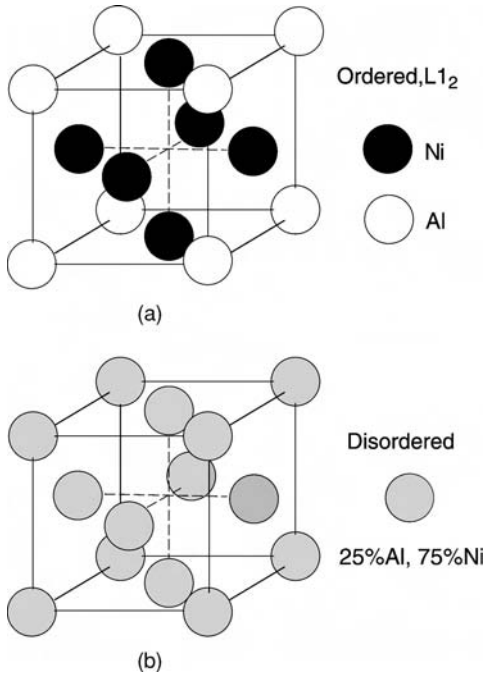


Fig. 2.11. Arrangements of Ni and Al atoms in (a) the ordered Ni_3Al phase and (b) after disordering.

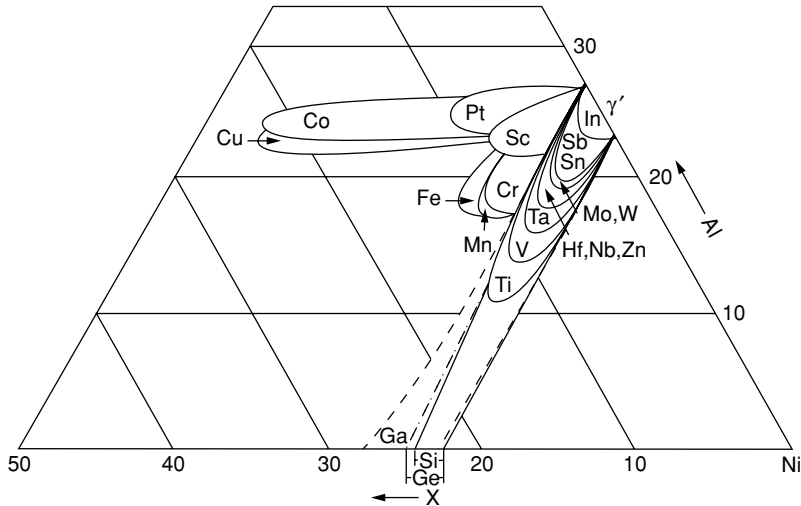


Fig. 2.12. Superimposed ternary phase diagrams Ni–Al–X [9], illustrating the great effect of X on the extent of the γ' phase field.

a constant Al fraction and providing confirmation that substitution for Ni on the first of the two sublattices is preferred. Elements such as Ti and Ta promote fields parallel to the Al–X axis, and thus they replace Al on the second sublattice. Only rarely, for example, for Cr, Fe and Mn, is mixed behaviour observed. The behaviour of the element X in this regard depends rather strongly on its size relative to Ni and Al. The lattice parameter, a , of γ'

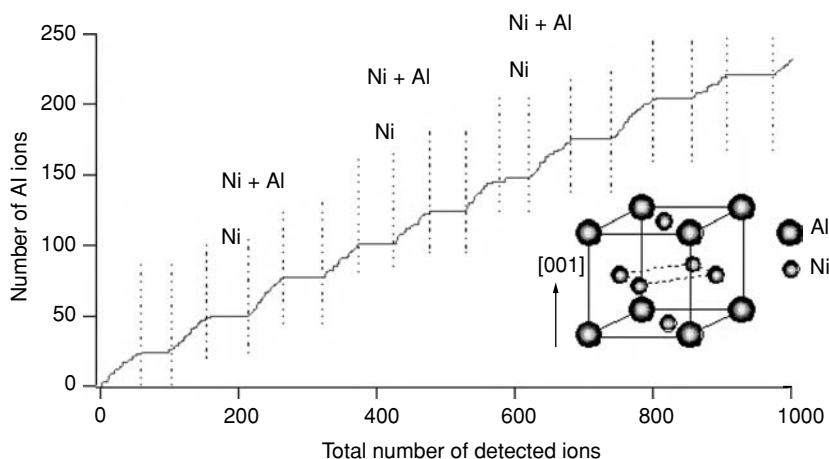


Fig. 2.13. Ladder diagram taken from a $\langle 001 \rangle$ -orientated Ni_3Al single crystal using atom probe field ion microscopy. Courtesy of Kazuhiro Hono.

at room temperature is 0.3570 nm, which is equivalent to the Al–Al distance; this is only $\sim 1.5\%$ larger than the lattice parameter of pure Ni, which is 0.3517 nm. The Ni–Al distance is $a/\sqrt{2}$, or 0.2524 nm. Thus substitution for Al is favoured by large elements such as Ta and Ti, whereas smaller atoms such as Co substitute for Ni.

Direct evidence of the ordered nature of γ' comes from experiments carried out using the atom probe, which consists of a field ion microscope coupled to a mass spectrometer. In these experiments, atoms are evaporated in the form of ions from a specimen sharpened to a radius of a few nanometres; a channel plate is used to select an appropriate pole, and the atoms are identified by measuring the time-of-flight to the detector, which depends uniquely on the charge-to-mass ratio. The sequential stripping of atoms from a series of $\{200\}$ planes allows a ladder diagram to be built up [10] (see Figure 2.13), on which the total number of Al ions detected during the run is plotted against the total number of ions collected. The mean gradient of the line is one-quarter, but during the stripping of mixed Ni/Al and Ni planes the gradient changes to one-half and zero, respectively. These features of the ladder diagram provide very elegant and direct proof of the ordering of the γ' phase on the atomic scale.

Does the γ' remain ordered until the melting temperature is reached? Disordering is a possibility which should be considered as the entropy contribution to the Gibbs energy becomes larger as the temperature is raised. For many years this question remained unanswered, but it is now known that the ordering temperature of Ni_3Al is roughly equivalent to its melting temperature of about 1375°C , although it does depend strongly upon the degree of stoichiometry and the concentrations of any impurities. This has been demonstrated [11] by doping Ni_3Al with varying amounts of Fe, which weakens the ordering; this lowers the critical ordering temperature, which can then be measured using dilatometry (see Figure 2.14). The ordering temperature of pure stoichiometric Ni_3Al is then estimated by extrapolation of the data to zero iron content; the result of $\sim 1375^\circ\text{C}$ must be termed ‘virtual’ since it lies above the melting point. In fact, evidence of the disordering of Ni_3Al

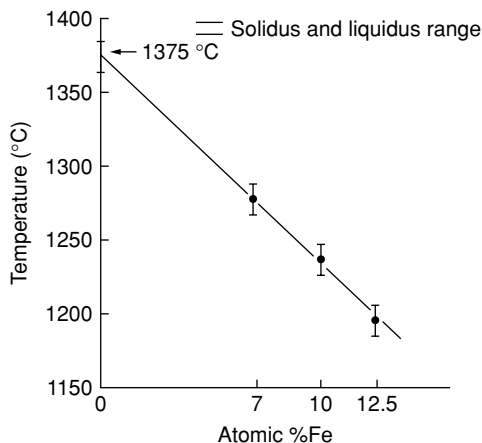


Fig. 2.14. Variation of the ordering temperature of Ni_3Al when doped with Fe [11]. Note that the ordering temperature of pure Ni_3Al can be deduced by extrapolation to zero Fe content.

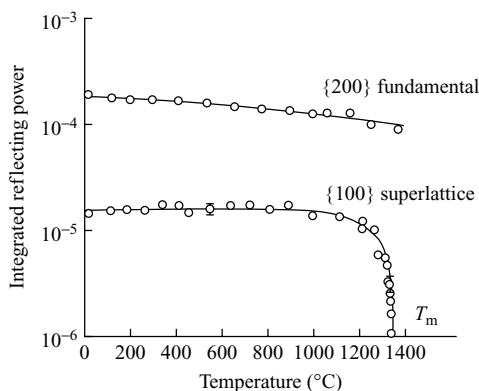


Fig. 2.15. Temperature dependence of the intensity of two diffraction lines from a Ni_3Al crystal, containing slightly less than 25 at%Al [12].

has been observed [12] using high-temperature neutron diffractometry; see Figure 2.15. In the vicinity of the melting temperature, the intensity of the $\{100\}$ superlattice reflection drops sharply – implying the onset of disordering before the melting temperature is reached.

Binary Ni–Al alloys of composition consistent with a two-phase γ/γ' microstructure exhibit γ' precipitates which are often cuboidal in form [13] (see Figure 2.16). Analysis of these structures using transmission electron microscopy confirms that a distinct cube–cube orientation relationship exists between the γ' precipitates and γ matrix in which they reside (see Figure 2.17). Since the lattice parameters of the disordered γ and ordered γ' phases are very similar, the electron diffraction patterns exhibit maxima which are common, for example, from the $\{110\}$, $\{200\}$, $\{220\}$ reflections, and others which are due only to γ' , for example, $\{100\}$, $\{210\}$ etc. The orientation relationship can be described by

$$\begin{aligned} \{100\}_{\gamma} // \{100\}_{\gamma'} \\ \langle 010 \rangle_{\gamma} // \langle 010 \rangle_{\gamma'} \end{aligned} \quad (2.5)$$

which is referred to as the cube–cube orientation relationship. The γ/γ' interfaces have the $\langle 001 \rangle$ directions as their plane normals. Provided that the lattice misfit between the lattice

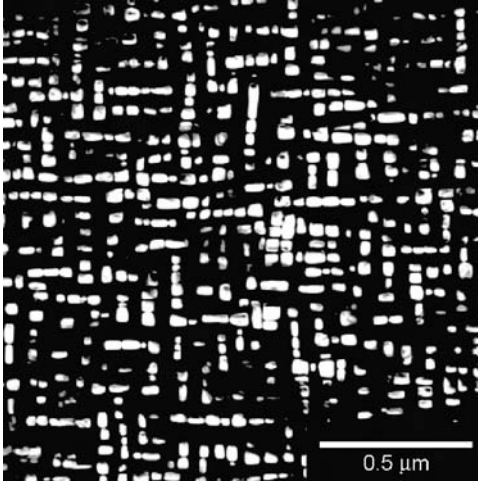


Fig. 2.16. Dark-field transmission electron micrograph of γ' precipitates in a Ni–13.4 at%Al alloy, aged at 640 °C for 1000 h [13].

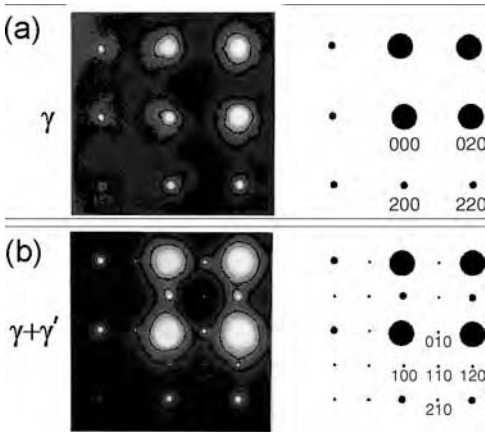


Fig. 2.17. Selected area diffraction patterns taken in the transmission electron microscope of a single-crystal superalloy, with foil normal close to $\langle 001 \rangle$, (a) of the disordered γ and (b) of both the ordered γ and disordered γ' , with superlattice reflections.

parameters of the γ and γ' phases is not too large, the γ/γ' interface remains coherent and the interfacial energy remains low. The γ' precipitates often align along the elastically soft $\langle 100 \rangle$ direction (see Figure 2.16).

The properties of the superalloys are found to depend critically on the coherency of the γ/γ' interface. This is favoured by small values of the lattice misfit, δ , defined according to

$$\delta = 2 \times \left[\frac{a_{\gamma'} - a_{\gamma}}{a_{\gamma'} + a_{\gamma}} \right] \quad (2.6)$$

which is thus negative if $a_{\gamma'} < a_{\gamma}$ and positive if the converse is true. Consistent with Vegard's law, one has relationships such as $a_{\gamma} = a_{\gamma}^0 + \sum_i \Gamma_i^{\gamma} x_i^{\gamma}$ and $a_{\gamma'} = a_{\gamma'}^0 + \sum_i \Gamma_i^{\gamma'} x_i^{\gamma'}$ so that the lattice parameters are linear with the mole fractions of added solutes. The Vegard coefficients Γ_i^{γ} and $\Gamma_i^{\gamma'}$ show a strong dependence on the position of i in the periodic table (see Figure 2.18). Values are largest for elements from the far west and far east of

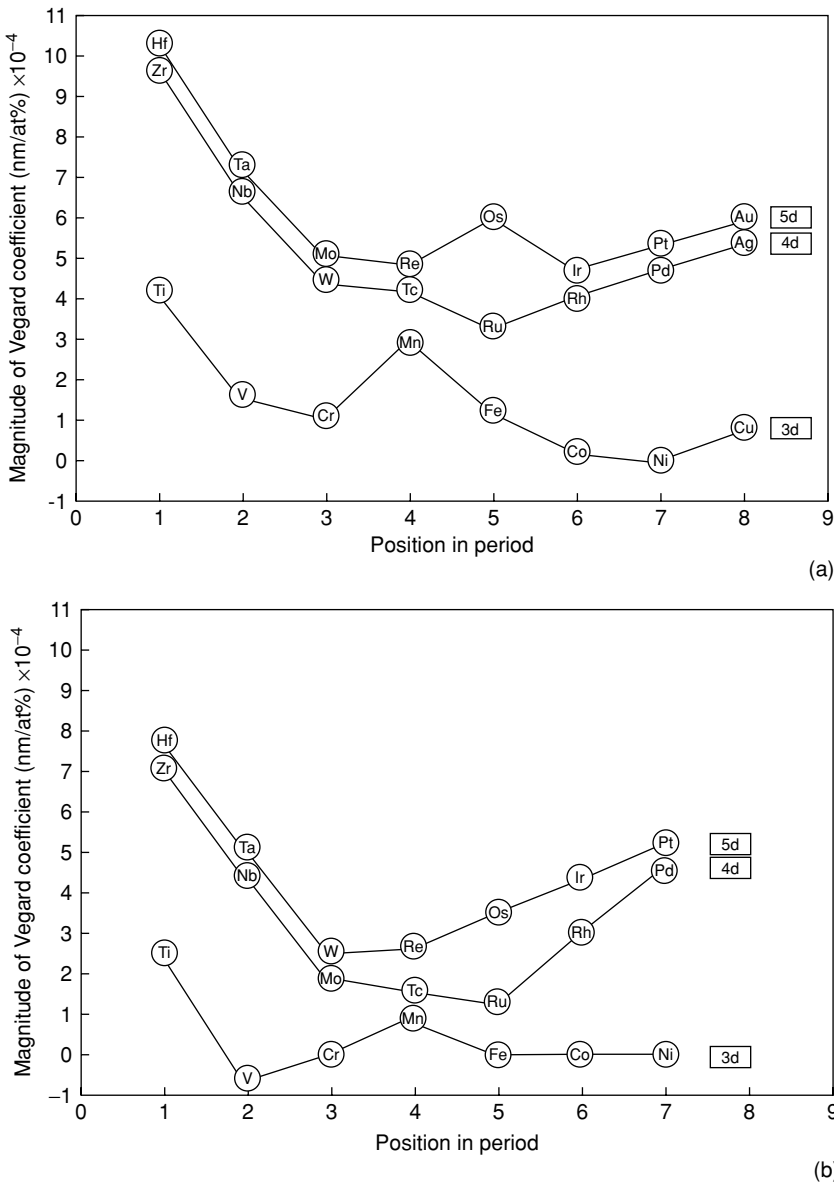


Fig. 2.18. Variation of the Vegard coefficients with position in the transition metal series: (a) for the disordered FCC phase, γ , and (b) for the ordered Ni_3Al phase, γ' . Data from ref. [3].

the transition metal block, when the discrepancy in atomic size is largest. Moreover, the lattice parameter of the γ phase shows the greater sensitivity to solute additions. The lattice misfit, δ , then depends on two factors: (i) the partitioning of solutes i between γ and γ' and (ii) the corresponding influence of the solutes i on the lattice parameters consistent with Vegard's law. Coarsening of the γ' particles leads to loss of coherency and an increase

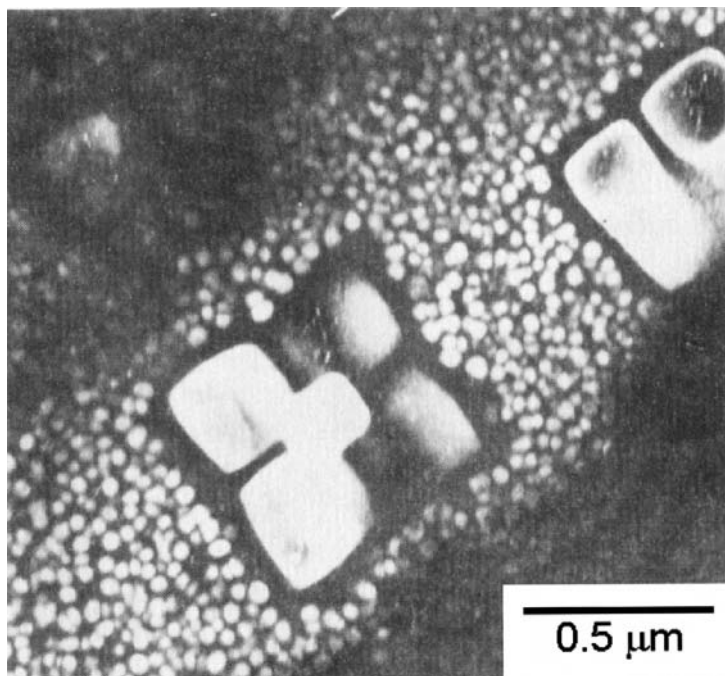


Fig. 2.19. Transmission electron microscope (TEM) image [14] showing γ' precipitation in Nimonic 115 aged at 1418 K for 24 h. Note (i) the central, initial cuboid from which the primary γ' particle has developed, (ii) the precipitate-free zone around it and (iii) the fine spherical dispersion of γ' formed upon cooling.

in the magnitude of δ . Thus a distinction should be drawn between the *constrained* and *unconstrained* lattice misfit.

Ricks *et al.* [14] have studied the development of γ' precipitates in a number of different nickel-based superalloys during heat treatment, identifying correlations between the morphology, size and sign of the γ/γ' misfit (see Figure 2.19). It was shown that the morphological development occurs in the sequence spheres, cubes, arrays of cubes and eventually solid-state dendrites as coarsening is promoted by ageing (see Figure 2.20). No substantial difference in this respect was found between positive and negative misfitting alloys, although in the latter case the $\{001\}$ facets were better defined and interfacial dislocation networks were found to develop at an earlier stage in the γ' growth sequence. Interestingly, both the size at which the γ' particles depart from the spherical morphology and the severity of the heat treatment required to form the cuboidal arrays were found to be sensitive functions of the lattice misfit; thus, for alloys of low misfit the size of the γ' particle needs to be larger before the effects of misfit strain are sufficient to influence the particle shape. For the Nimonic alloys 80A, 90, 105 and 115, for which convergent beam electron diffraction (CBED) was used to estimate the room-temperature misfits at +0.32%, +0.34%, -0.04% and -0.18%, respectively, the γ' precipitate sizes at which significant departure from the spherical shape was observed were found in turn to be 0.3, 0.3, 0.7 and

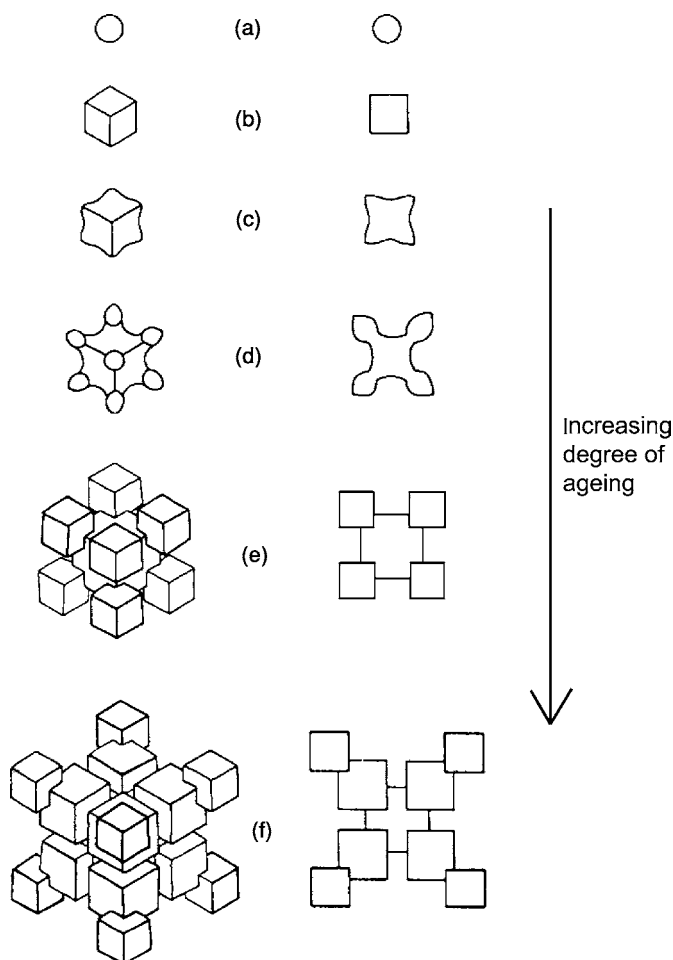


Fig. 2.20. Schematic diagram illustrating the development, from (a) to (f), of strain-induced, faceted γ' morphologies during ageing: left, projection along $\langle 111 \rangle$; right, projection along $\langle 001 \rangle$. Adapted from ref. [14].

0.5 μm . This suggests that when the magnitude of the misfit is small, the γ' particles must grow to a larger size before the cuboidal form is found.

2.1.3 Other phases in the superalloys

A The gamma double prime phase

In nickel–iron superalloys such as IN718 and IN706, which contain quantities of niobium, the primary strengthening precipitate is not γ' but instead a body-centred tetragonal (BCT) ordered compound; this displays the $D0_{22}$ crystal structure and a composition which can be represented approximately by Ni_3Nb . In IN718, the lattice parameters are approximately $c = 0.740 \text{ nm}$ and $a = 0.362 \text{ nm}$ [15]. The unit cell is shown in Figure 2.21. One can see

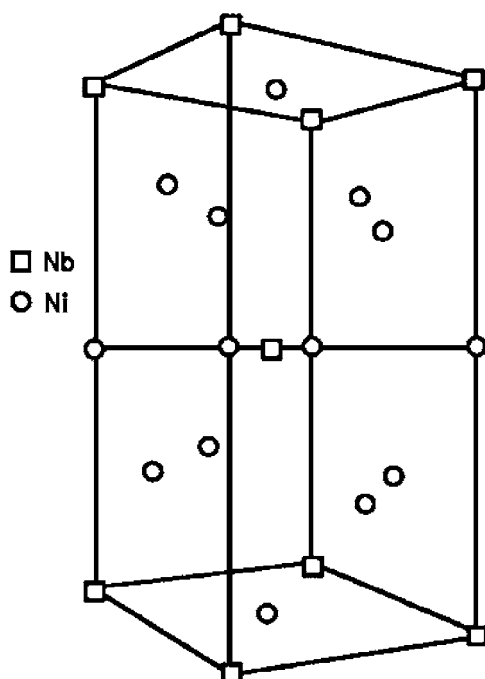


Fig. 2.21. The unit cell of the γ'' precipitate which is found in iron–nickel superalloys such as IN718 and IN706.

that the arrangement of the atoms is related to that of the $L1_2$ structure of the γ' phase; in fact the a parameter is close to that displayed by γ' but the c parameter is roughly doubled. Because of this similarity with γ' , one refers to this phase as γ'' . As with γ' , a distinct orientation relationship with the matrix phase γ is displayed, such that $\langle 001 \rangle_{\gamma''} // \langle 001 \rangle_{\gamma}$ and $\langle 100 \rangle_{\gamma''} // \langle 100 \rangle_{\gamma}$. The γ'' displays a disc-shaped morphology, with the thickness often as small as 10 nm and the diameter about 50 nm (see Figure 2.22). The particles are usually found to be coherent with the matrix, with coherency strains equivalent to several per cent. The excellent high-temperature properties of IN718 – which is used very widely in polycrystalline cast-and-wrought form to operating temperatures approaching 650 °C on account of its relatively low cost – are due to the coherency strains so imparted and the limited number of available slip systems which operate in γ'' . The kinetics of formation of γ'' are sluggish as a consequence of the high coherency strains.

Nickel–iron alloys which are strengthened by γ'' are susceptible to the formation of an orthorhombic δ phase in the overaged condition [16,17]. This is invariably incoherent with γ , and therefore does not confer strength even when present in significant quantities. It forms in the temperature range 650 to 980 °C, but the characteristics of its formation depend strongly upon temperature. Below 700 °C, nucleation of δ is observed at γ –grain boundaries and growth occurs at the expense of γ'' . In the range 700 °C to 885 °C, formation of δ is accompanied by rapid coarsening of γ'' ; beyond the solvus temperature of 885 °C γ'' is no longer stable. At temperatures between 840 °C and 950 °C, the plates of δ form rapidly in times less than 24 h. The δ solvus temperature is ~ 1000 °C. It should be emphasised, however, that the kinetics of formation of δ are strongly accelerated if forging is carried

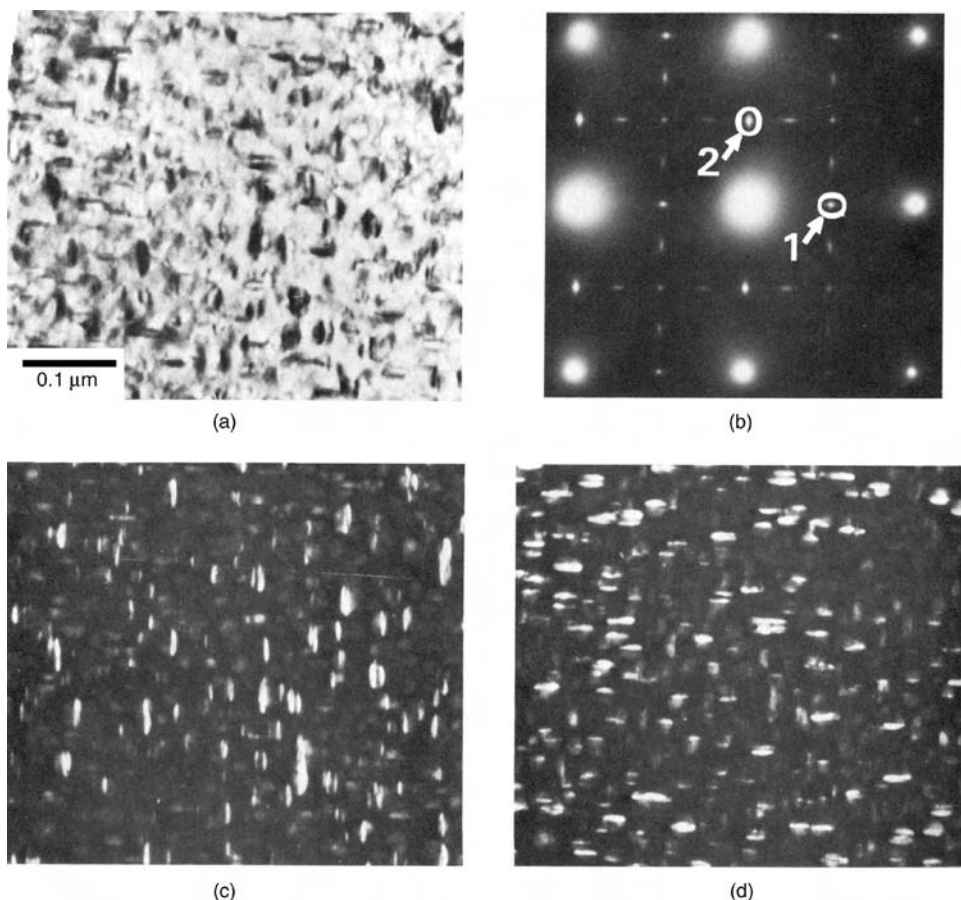


Fig. 2.22. Transmission electron micrographs [17] showing the γ' and γ'' phases in the IN706 alloy aged at 732 °C for 16 h: (a) bright-field image, (b) diffraction pattern, (c) dark-field image using spot 1 and (d) dark-field image using spot 2.

out below the δ solvus; the resultant δ -phase distribution can be used to control and refine the grain size to help optimise the tensile and fatigue properties. However, the formation of extensive amounts of δ during service leads to severe degradation of properties and is to be avoided.

B The TCP phases

Excessive quantities of Cr, Mo, W and Re promote the precipitation of intermetallic phases which are rich in these elements [18] (see Figure 2.23). The resulting phases have a number of distinct characteristics: (i) a high and uniform packing density of atoms, (ii) a degree of non-metallic, directional bonding and (iii) complex crystal structures, each built up of distinct tessellated layers consisting of arrays of hexagons, pentagons and triangles (see Figure 2.24) stacked into a limited number of so-called Kasper co-ordination polyhedra

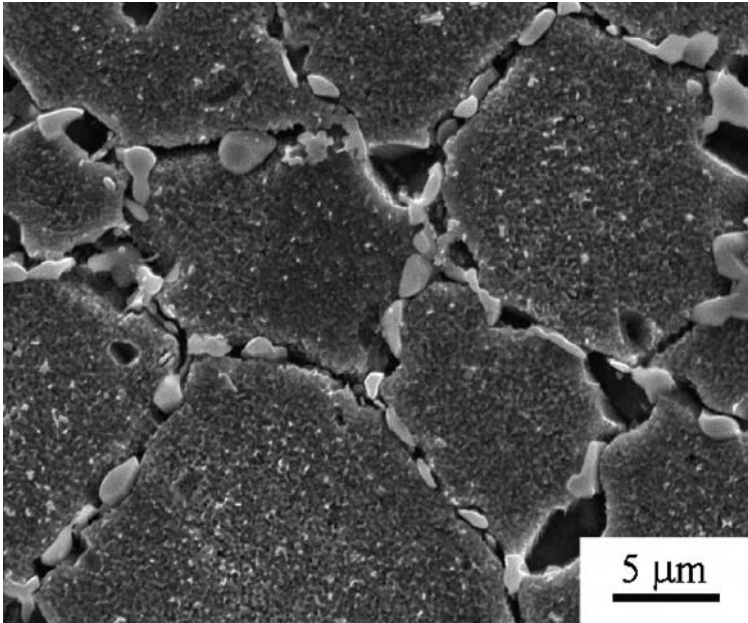


Fig. 2.23. Scanning electron micrograph of the superalloy RR1000 showing extensive precipitation of the σ phase at γ -grain boundaries, after a heat treatment of 5000 h at 750 °C. Courtesy of Rob Mitchell.

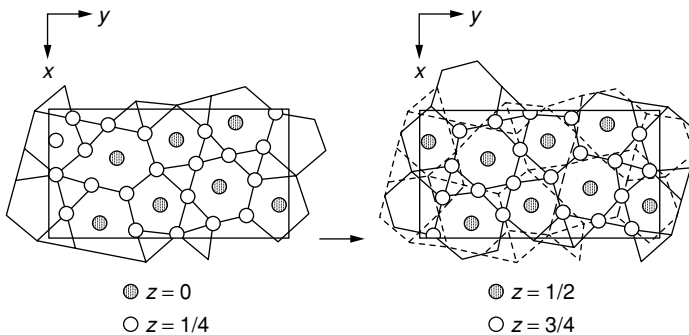


Fig. 2.24. The unit cell of the P phase showing the primary and secondary layers. For clarity, all atoms are represented by circles of the same size. Adapted from ref. [19].

[19]. This last fact gives rise to the name *topologically close-packed*, or TCP, phases. Generally speaking, the TCP phases have chemical formulae A_xB_y , where A and B are transition metals, such that A falls to one side of the group VIIB column defined by Mn, Tc and Re (see Figure 1.21) and B falls to the other [19]. The μ phase is based on the ideal stoichiometry A_6B_7 and has a rhombohedral cell containing 13 atoms; examples include W_6Co_7 and Mo_6Co_7 . The σ phase is based upon the stoichiometry A_2B and has a tetragonal cell containing 30 atoms; examples include Cr_2Ru , $Cr_{61}Co_{39}$ and $Re_{67}Mo_{33}$. The P phase, for example, $Cr_{18}Mo_{42}Ni_{40}$ is primitive orthorhombic, containing 56 atoms per cell. Finally,

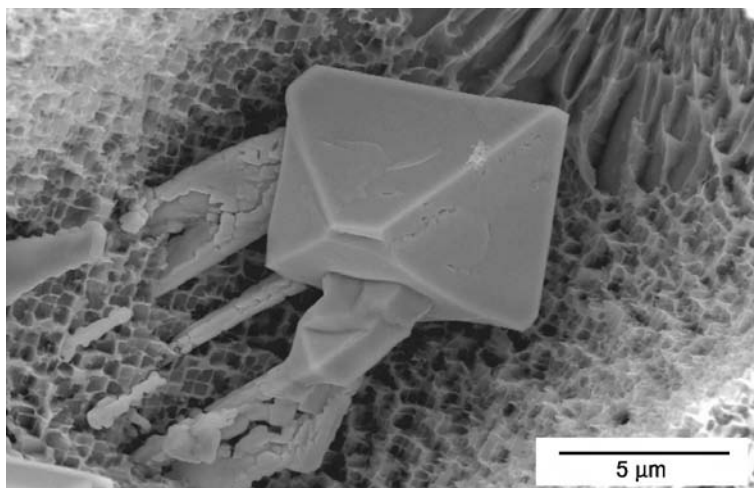


Fig. 2.25. Example of a blocky MC carbide formed in an experimental nickel-based single-crystal superalloy [20].

the R phase, for example, $\text{Fe}_{52}\text{Mn}_{16}\text{Mo}_{32}$, has a rhombohedral cell containing 53 atoms. It should be emphasised that the TCP phases can have very different compositions from those quoted here, and that the stoichiometry ranges are often very wide. However, the fact that the crystal structures are based upon A_xB_y , with A and B thus displaying different electronegativities, suggests that electronic factors are important in determining stability, in an analogous way to the ‘electronic compounds’ of Hume-Rothery in alloys which contain combinations of Cu, Ag and Au.

C Carbide and boride phases

Various carbide and boride species form in the superalloys, the type depending upon the alloy composition and the processing conditions employed. Some of the more important types include MC, M_6C , M_{23}C_6 , M_7C_3 and M_3B_2 , where M stands for a metal atom such as Cr, Mo, Ti, Ta or Hf. In many superalloys the MC carbide, which is usually rich in Ti, Ta and/or Hf since these are strong carbide formers, precipitates at high temperatures from the liquid phase. Consequently, the carbide is often found in interdendritic regions and no distinct orientation relationship with the matrix is displayed. A number of morphologies, for example, globular, blocky and script have been reported [20] (see Figure 2.25). Carbides such as M_{23}C_6 form at lower temperatures – around 750°C – during protracted periods of service exposure, particularly in alloys which are rich in Cr. Their formation has been attributed to the breakdown of the MC carbides, via reactions of the type



The M_{23}C_6 carbide is usually found to precipitate on the γ -grain boundaries. The complex cubic structure exhibited by this phase is, in fact, closely related to that of σ – with the carbon atoms removed, the crystal structures resemble each other [18]. This fact explains

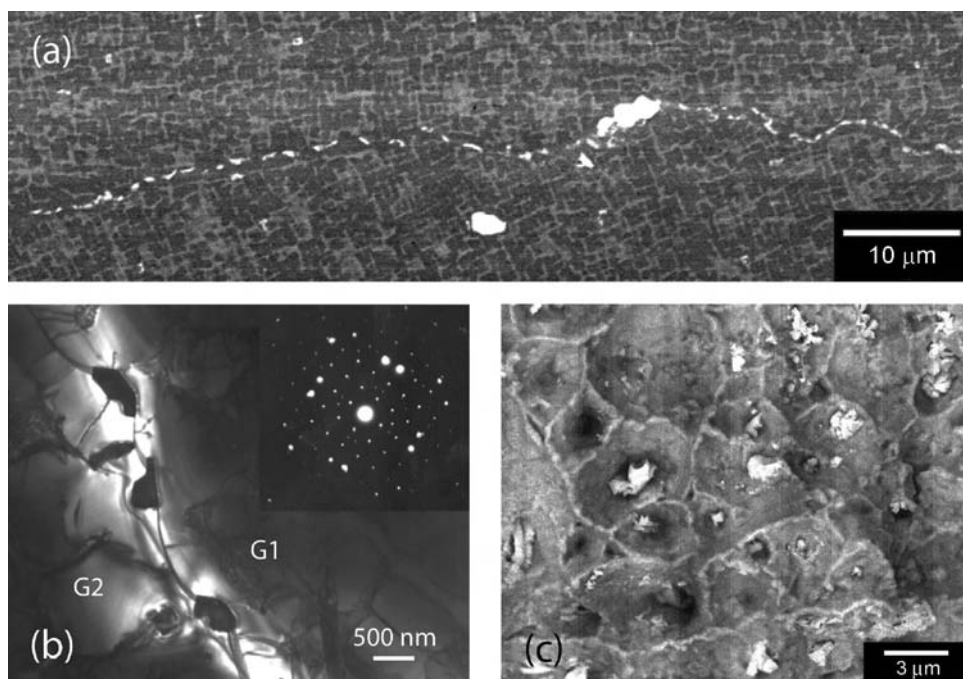


Fig. 2.26. Precipitation of the $M_{23}C_6$ carbide on the γ -grain boundary in an experimental second-generation single-crystal superalloy doped with carbon and boron and crept at 950°C and 290 MPa [21]: (a) scanning electron micrograph; (b) transmission electron micrograph and diffraction pattern identifying the $M_{23}C_6$ carbide; (c) dimpled fracture surface of creep specimen, showing evidence of ductile failure mode.

why σ is found to nucleate sympathetically from $M_{23}C_6$ in polycrystalline turbine disc alloys such as Udimet 720Li and RR1000.

The role of carbides and borides has proved controversial, but it is now accepted that the high-temperature creep properties of the superalloys are improved in many circumstances if carbon and boron are present; the formation of these phases is then very likely. This is primarily due to the preferred location of carbon, boron, carbides and borides at the γ -grain boundaries, which has a potent effect on the rupture strength via the inhibition of grain-boundary sliding. Thus, carbon and boron are often referred to as grain-boundary strengtheners. This explains why carbon and boron levels are generally higher in polycrystalline or columnar-grained superalloys used in cast form, and the absence of these elements in the single-crystal superalloys, for which grain-boundary strengthening is unnecessary. The benefits of doping with carbon and boron when grain boundaries are present has been confirmed by Chen and Knowles [21], who tested an experimental second-generation single-crystal superalloy in creep at 950°C and 290 MPa. When grain boundaries of misorientation equal to 10° were introduced, carbon and boron concentrations of 0.09 and 0.01 wt%, respectively, were sufficient to improve the rupture life from 10 h to 100 h. This effect was attributed to small, coherent and closely spaced $M_{23}C_6$ carbides precipitating on the grain boundaries (see Figure 2.26).

2.2 Defects in nickel and its alloys

A nickel alloy, despite being a crystalline solid, possesses a number of intrinsic defects present on the scale of the lattice. One can distinguish between (i) planar defects such as stacking faults, (ii) line defects such as dislocations and (iii) point defects, for example, vacancies, interstitials and anti-site defects. It turns out that the mechanical properties displayed by the superalloys are strongly dependent upon the type and concentration of defects which occur. In this section, consideration is given to the details of the defects which arise and to some of the implications of their presence.

2.2.1 Defects in the gamma (FCC) phase

The face-centred cubic (FCC) structure consists of close-packed planes, stacked with periodicity equal to three. Thus the lattice, if it is perfect, can be denoted ABCABCABCABC... , where each of A, B or C represents a close-packed layer; this contrasts with the hexagonally close-packed (HCP) structure, which is denoted ABABABABAB... . Since the lattice vector is $a/\sqrt{2}$, where a is the length of the side of the unit cell, one can demonstrate that two consecutive close-packed layers are displaced by a vector of magnitude $a/\sqrt{6}$, measured within a close-packed plane. The spacing of the close-packed planes is $a/\sqrt{3}$.

A Planar defects – the stacking fault

A stacking fault arises when the stacking sequence ABCABCABCABC... is interrupted [22]. The *removal* of one plane introduces an *intrinsic* stacking fault, for example, ABCACABCABC..., and causes two neighbouring planes to possess the HCP co-ordination locally. Conversely, the *addition* of one plane introduces an *extrinsic* stacking fault, for example, ABCACBCABCABC... . Although two HCP planes are again introduced, these are separated by a single plane of FCC co-ordination, and are thus a distance $2a/\sqrt{3}$ apart – a spacing corresponding to two lattice planes.

B Line defects – dislocations

In a macroscopic sense, the slip system in an FCC metal such as Ni is $a/2\langle 1\bar{1}0\rangle\{111\}$; hence the Burgers vector is $a/\sqrt{2}$. However, this statement does not properly respect the micromechanics of deformation; it is well established that the glide of an $a/2\langle 1\bar{1}0\rangle\{111\}$ dislocation occurs by the passage of two *partial* dislocations, which, although in close proximity to each other, are separated by a distance which depends upon the force of their elastic repulsion and the energy of the planar stacking fault so produced (see Figure 2.27). This implies that the $a/2\langle 1\bar{1}0\rangle\{111\}$ dislocations are dissociated; electron microscopy confirms that the reaction is of the form

$$\frac{a}{2}\langle 110\rangle\{\bar{1}11\} \rightarrow \frac{a}{6}\langle 211\rangle\{\bar{1}11\} + \frac{a}{6}\langle 12\bar{1}\rangle\{\bar{1}11\} \quad (2.8)$$

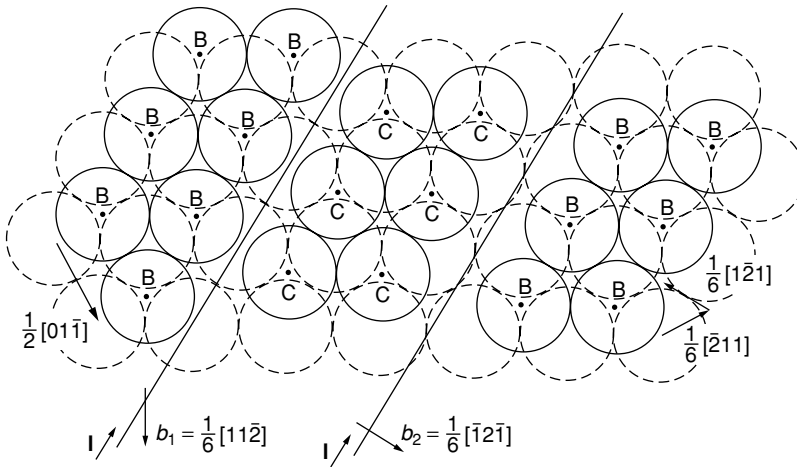


Fig. 2.27. Dislocation on a $\{111\}$ plane of γ , dissociated into two Shockley partial dislocations [22].

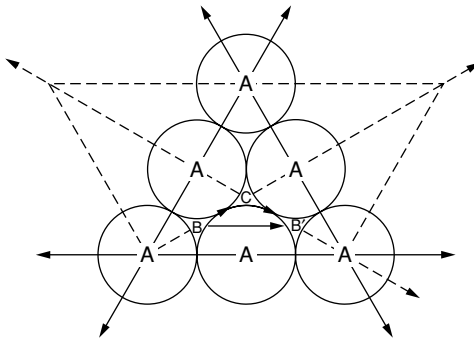


Fig. 2.28. Geometry of the close-packed plane, appropriate for the dissociation into Shockley partials; the large arrow corresponds to $\frac{a}{2}\langle 110 \rangle$, and the smaller arrows correspond to $\frac{a}{6}\langle 211 \rangle$ and $\frac{a}{6}\langle 121 \rangle$.

where each dislocation on the right-hand side of the equation is known as a Shockley partial. The magnitude of the Burgers vector of each is $a/\sqrt{6}$, equivalent to the displacement of two neighbouring close-packed planes in the perfect FCC crystal (see Figure 2.28). Thus the passage of a single Shockley partial dislocation causes the introduction of an intrinsic stacking fault, which is removed by the action of the second – hence the width of the stacking fault depends upon the spacing of the partials. The reader can readily apply Frank's rule, which states that the elastic energy of a dislocation is proportional to the square of the magnitude of the Burgers vector, to confirm that the dissociation of Equation (2.8) is energetically favourable.

It is of interest to determine the relationship between the spacing of the partial dislocations and the stacking fault energy (SFE) – the energy per unit area of the planar defect between the two partials, denoted γ_{SF} . Consider the dissociation of an edge dislocation (Figure 2.29). The spacing of the partials is given the symbol d ; each has a Burgers vector of magnitude $a/\sqrt{6}$, which can be resolved into an edge component, $a \cos\{\pi/6\}/\sqrt{6}$, and a screw component, $a \sin\{\pi/6\}/\sqrt{6}$, $\pi/6$ being the angle between the Burgers vectors of the dissociated and undissociated dislocations. It follows from the geometry of the situation, see Figure 2.29,

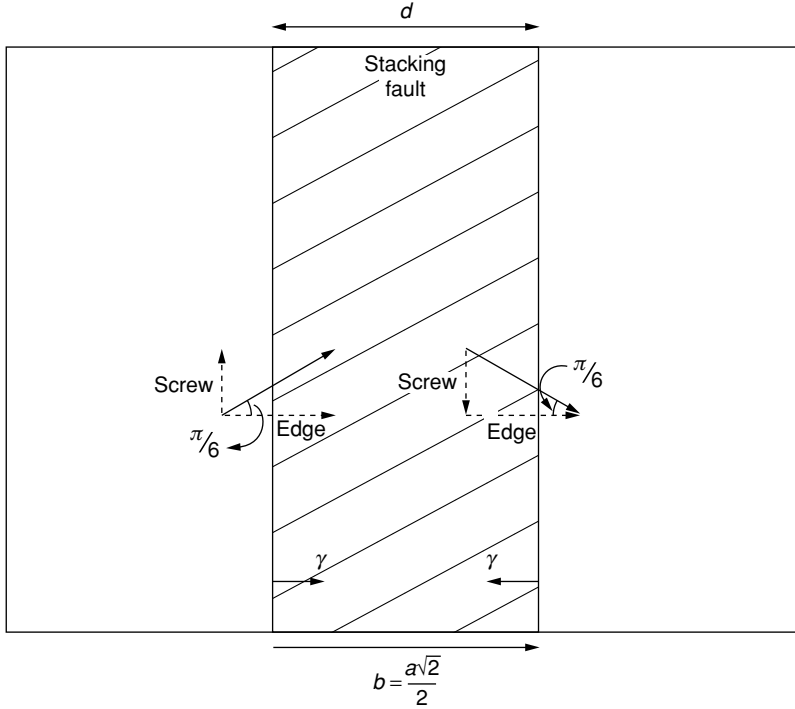


Fig. 2.29. Schematic illustration of the dissociation of an edge dislocation into two Shockley partial dislocations, which are in turn resolved into their edge and screw components.

that the screw components of the dissociated dislocations are of opposite sense so that they provide an attractive force complementing that due to the stacking fault energy. On the other hand, the edge components are of equivalent sign and thus repel each other; it is this effect which provides the driving force for dissociation.

The force balance consists therefore of three terms, the repulsive force per unit length of dislocation due to the edge components of the partials, balanced at equilibrium by attractive forces per unit length arising from the screw components and the stacking fault energy itself. It is given by

$$\gamma_{\text{SF}} + \frac{Gb_s^2}{2\pi d} = \frac{Gb_e^2}{2\pi d(1-\nu)} \quad (2.9)$$

where b_e and b_s are the edge and screw components of the partials of magnitude $a \cos\{\pi/6\}/\sqrt{6}$ and $a \sin\{\pi/6\}/\sqrt{6}$, respectively, G is the shear modulus and ν is Poisson's ratio. Note that isotropic elasticity is assumed to hold. The forces $Gb_s^2/2\pi d$ and $Gb_e^2/2\pi d(1-\nu)$ arise from the shear stresses $Gb_s/2\pi d$ and $Gb_e/2\pi d(1-\nu)$ acting at a distance d from the centres of perfect screw and edge dislocations, respectively, as given by elementary dislocation theory; the forces per unit length, F , are obtained by multiplying by the appropriate shear stress, τ , by the corresponding Burgers vector, b , consistent with $F = \tau b$. After some algebraic manipulation, one can show that the spacing of the partials

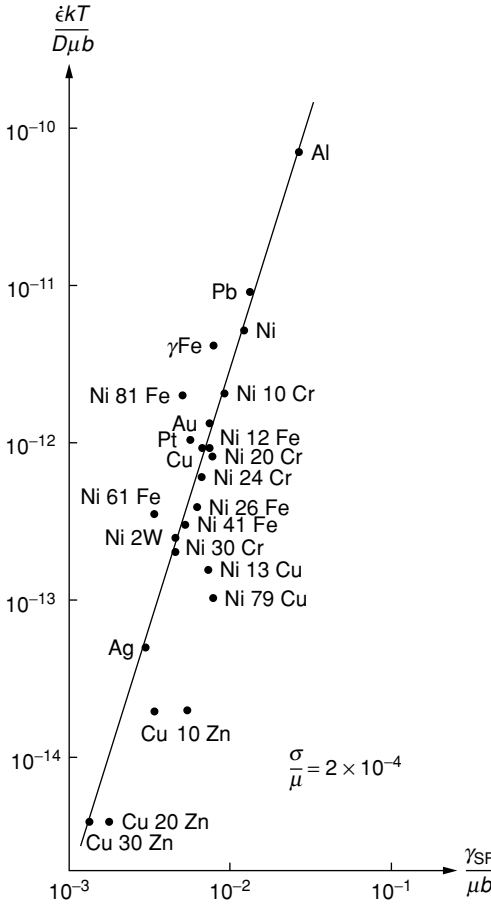


Fig. 2.30. Correlation between the normalised steady-state creep rate, $\dot{\epsilon}$, and the stacking fault energy, γ_{SF} , for a number of solid-solution-strengthened alloys [23].

is given by

$$d = \frac{Ga^2}{48\pi\gamma_{SF}} \left(\frac{2 + \nu}{1 - \nu} \right) \quad (2.10)$$

the important result being that the spacing, d , is inversely proportional to the stacking fault energy, γ_{SF} . All dislocations, whether edge, screw or mixed, are dissociated in this way. It can be demonstrated by the reader that if a screw rather than an edge dislocation is considered, the spacing, d , is reduced by a factor $(2 - 3\nu)/(2 + \nu)$, which is approximately one-half if $\nu = 0.33$.

For high-temperature applications, the magnitude of the stacking fault energy is an important consideration. This is because screw portions of perfect dislocations, dissociated into their Shockley partials, must constrict locally if they are to cross-slip. This is a common mechanism by which obstacles such as precipitates can be circumvented. On this basis, widely spaced partials arising from low values of γ_{SF} are advantageous to high-temperature

properties; this has been confirmed by the work of Mohamed and Langdon [23], who measured the steady-state creep rate, $\dot{\epsilon}$, of a number of solid-solution-strengthened nickel alloys (see Figure 2.30). A correlation between the parameter $\dot{\epsilon}kT/(DGb)$ was found, where D is the diffusion coefficient, and the stacking fault energy, γ_{SF} , normalised by Gb . Transmission electron microscopy studies have confirmed a strong influence of alloying elements on the stacking fault energy. See, for example, the data [24] for Ni–Co alloys (Figure 2.31), which indicates that γ_{SF} decreases from about 250 mJ/m² for pure Ni to less than half this value for an alloy of composition Ni–50wt%Co. It appears, however, that there have been no systematic attempts to correlate the influence of stacking fault energy on the type of solute, although Delehouzee and Deruyttere [25] have measured the stacking fault probability, α , a measure of the number of stacking faults existing in a metal, from the broadening and shifting of diffraction peaks using X-ray methods; this is expected to be inversely proportional to γ_{SF} . Their results are given in Figure 2.32. The slope of the lines correlate strongly with the valency difference between Ni and the solute being added. Given these results, on the basis of at% solute added one would expect elements such as Ti, which lie at the far left of the d block of transition metals, to have the greatest effect on γ_{SF} and the high-temperature properties which are influenced by it. Such alloying effects are important since Ni itself has one of the largest stacking fault energies of the FCC metals at 250 mJ/m²; by contrast, for Al, Cu, Au and Ag the values are approximately 200, 55, 50 and 22 mJ/m², respectively. Thus the partial dislocations in unalloyed Ni are relatively closely spaced.

C Point defects – vacancies

Vacancies are present in the lattice of high-purity Ni, because the increase in free energy caused by the breaking of bonds is more than offset by the term due to the configurational entropy which then arises. From statistical mechanics one expects the equilibrium vacancy fraction, $f_{\text{Va}}^{\text{eq}}$, to be given by

$$f_{\text{Va}}^{\text{eq}} = \exp \left\{ -\frac{\Delta G_{\text{Va}}^f}{RT} \right\} = \exp \left\{ +\frac{\Delta S_{\text{Va}}^f}{R} \right\} \times \exp \left\{ -\frac{\Delta H_{\text{Va}}^f}{RT} \right\} \quad (2.11)$$

where ΔG_{Va}^f , ΔH_{Va}^f and ΔS_{Va}^f are, respectively, the Gibbs energy, enthalpy and entropy of formation of a vacancy. For most metals, the pre-exponential term given by $\exp \{ +\Delta S_{\text{Va}}^f/R \}$ lies between 1 and 10, and is nearly invariant with temperature. Electrical resistivity measurements on quenched and quenched/tempered nickel [26] indicate that ΔH_{Va}^f is ~ 160 kJ/mol; taking the pre-exponential term as 5, this gives a fraction f of 10^{-28} at ambient temperature, rising to 8.5×10^{-5} at the melting temperature of 1455 °C. This value is comparable to other FCC metals; for example, for Al, Au and Ag the enthalpies ΔH_{Va}^f are 65 kJ/mol, 85 kJ/mol and 107 kJ/mol, respectively, so that $f_{\text{Va}}^{\text{eq}}$ is 9.4×10^{-4} , 7.2×10^{-5} and 1.7×10^{-5} at the melting temperature of the metal concerned.

The presence of vacancies can be inferred directly from measurements of the lattice spacing and thermal expansion coefficient, i.e. using a combination of diffractometry and dilatometry. This is because one atomic site is added each time a vacancy is created at a free surface, grain boundary or dislocation. The corresponding changes in the dimensions of the

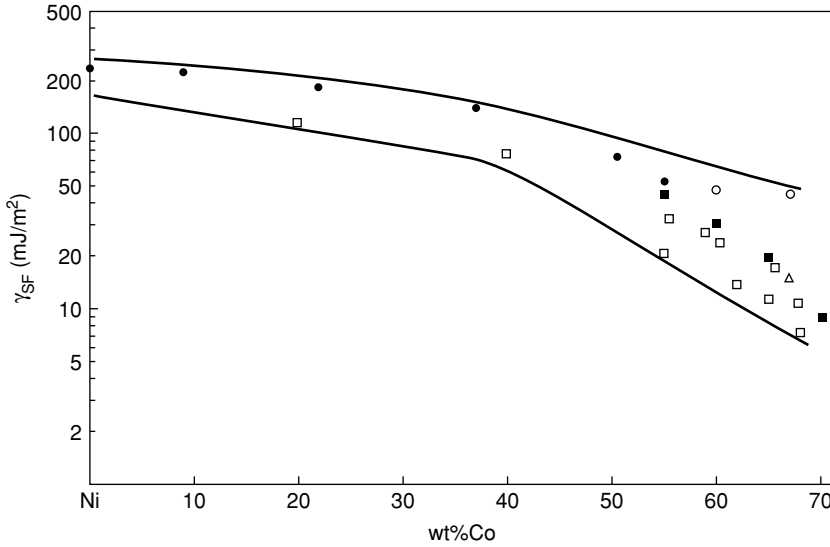


Fig. 2.31. Variation of the stacking fault energy, γ_{SF} , with composition for a number of binary Ni-Co alloys [24].

specimen will be detectable by dilatometry; however, if internal stresses can be ignored, the lattice spacing will not be altered substantially. To illustrate this, consider a one-dimensional analogy: a perfect row of N atoms of spacing a_0 which is heated from T_0 to T_1 , causing (i) the formation of n vacancies such that $n \ll N$ and (ii) a change in length from L_0 to L_1 . The expansion $\Delta L = L_1 - L_0$ has two contributions: the first is due to the increase in the periodicity from a_0 to a_1 due to the thermal expansion, denoted Δa , and the second is due to the introduction of the vacancies. Hence,

$$\Delta L = (N \times \Delta a) + (n \times a_1) \quad (2.12)$$

Dividing through by $Na_1 \sim L_1$ and rearranging, one has

$$f_{va}^{eq} = \frac{n}{N} = \frac{\Delta L}{L_1} - \frac{\Delta a}{a_1} \quad (2.13)$$

For metals, this simple estimate is, in fact, in error by a factor of 3, since one has in reality a three-dimensional rather than a one-dimensional lattice; the fraction f_{va}^{eq} is then given instead by $3(\Delta L/L_1 - \Delta a/a_1)$. Small errors are expected due to strain relaxation effects and the possibility of vacancy clustering to form divacancies; nevertheless, $3(\Delta L/L_1 - \Delta a/a_1)$ is usually found to be of the order 10^{-4} for most metals [27]; see Figure 2.33.

The presence of vacancies in the nickel alloys is of great significance, since they mediate the diffusional flow required for a number of phenomena: creep occurring at a rate dependent upon diffusional rearrangements at dislocation cores, coarsening of the γ' precipitates on the scale of their periodicity, and oxidation, which occurs at a rate which is diffusion-controlled; clearly, the retardation of diffusional processes is important if the very best properties are to be attained. Consider the self-diffusion of nickel. The diffusion coefficient, D_{Ni}^{self} , depends on two factors:

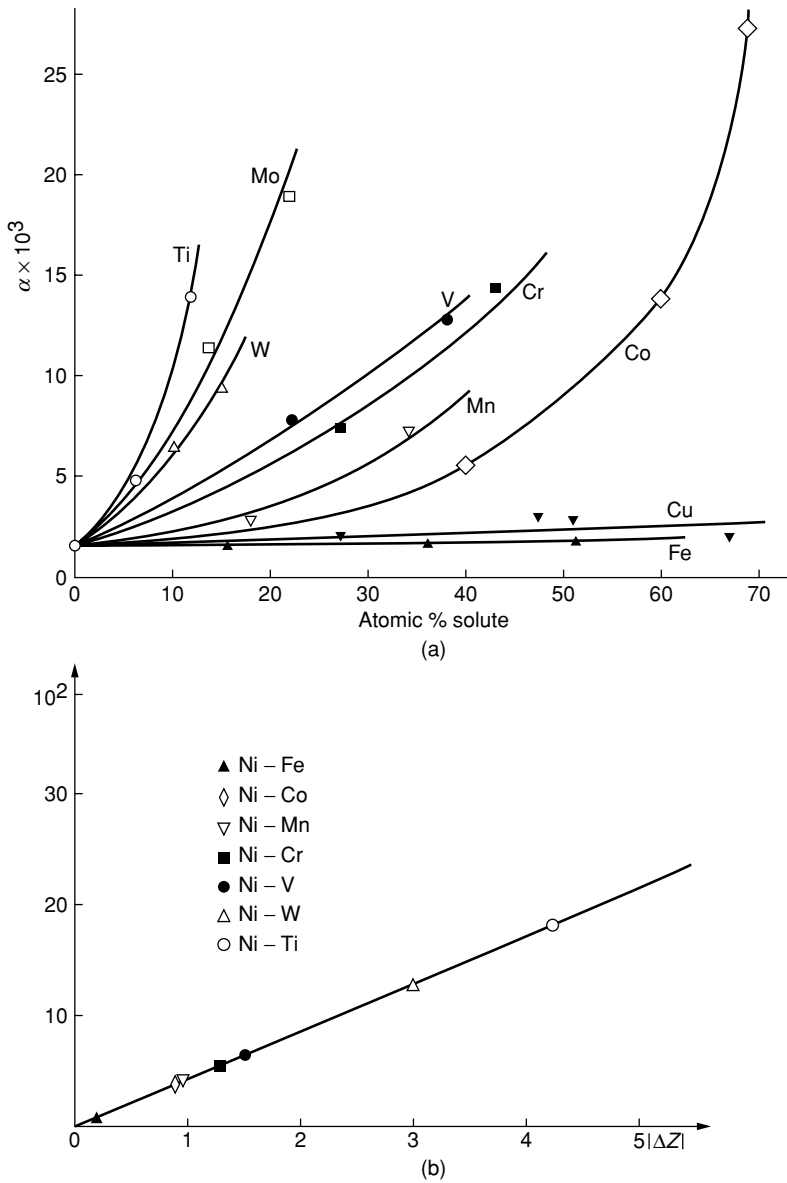


Fig. 2.32. For a number of binary nickel alloys [25]: (a) variation of the stacking fault probability with solute content, and (b) correlation of the slope of (a) at zero solute content, with the difference between the valency of nickel and solute denoted ΔZ .

(i) the rate of successful atom–vacancy exchanges, denoted Γ_{va} , which is given by

$$\Gamma_{va} = \nu_o \exp \left\{ -\frac{\Delta E_{Ni-Va}^{mig}}{RT} \right\} \quad (2.14)$$

where ν_o is an attempt frequency and ΔE_{Ni-Va}^{mig} is an activation energy associated with

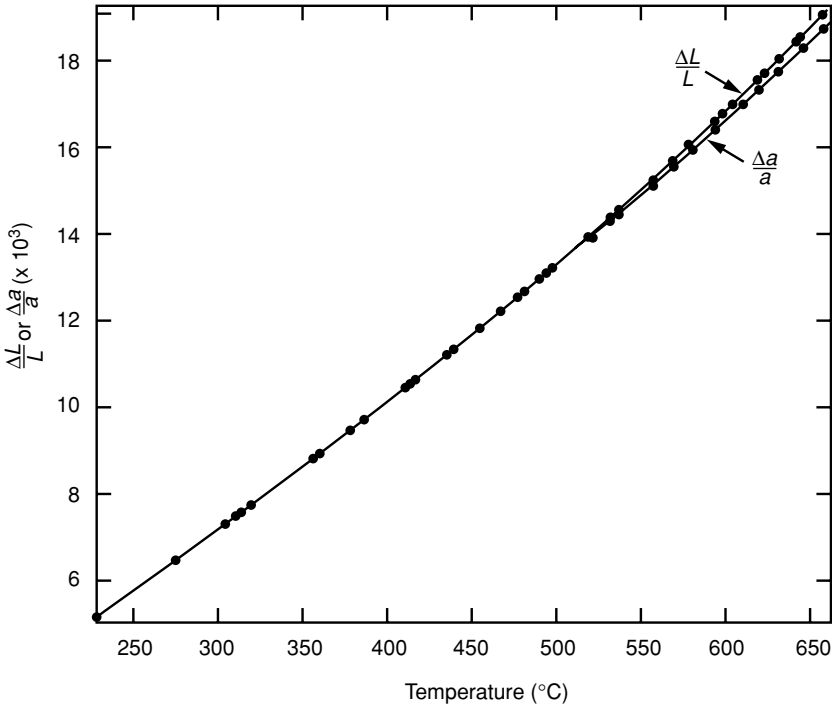


Fig. 2.33. Effects of changes in length and lattice parameter with temperature [27].

the migration of a nickel atom into a neighbouring vacant site, through a distance $a/\sqrt{2}$ along the lattice vector $\langle 110 \rangle$, past the saddle point;

- (ii) the probability that the neighbouring site is vacant. This is numerically equivalent to the fractional concentration of lattice vacancies in thermal equilibrium with the lattice, as given by Equation (2.11).

One has therefore, to a first approximation,

$$D_{\text{Ni}}^{\text{self}} = \lambda^2 \nu_o \exp \left\{ + \frac{\Delta S_{\text{Va}}^f + \Delta S_{\text{Ni-Va}}^{\text{mig}}}{R} \right\} \times \exp \left\{ - \frac{\Delta H_{\text{Va}}^f + \Delta H_{\text{Ni-Va}}^{\text{mig}}}{RT} \right\} \quad (2.15)$$

where the activation energy, ΔE_{mig} , has been partitioned into its enthalpy and entropy components, ΔH_{mig} and ΔS_{mig} , respectively. The term λ is the characteristic jump distance given by $a/\sqrt{2}$. One can see that the apparent activation energy for self-diffusion, Q_{self} , is composed of two terms, such that

$$Q_{\text{self}} = \Delta H_{\text{Va}}^f + \Delta H_{\text{Ni-Va}}^{\text{mig}} \quad (2.16)$$

The experimental measurements indicate that $Q_{\text{self}} \sim 280 \text{ kJ/mol}$ [28]. Hence, $\Delta H_{\text{Ni-Va}}^{\text{mig}} \sim 120 \text{ kJ/mol}$, given the value of ΔH_{Va}^f quoted above. This value is associated with the opening of the diffusion window along $\langle 110 \rangle$.

The presence of vacancies influences the rate of diffusion of substitutional solutes in nickel. The rate of interdiffusion of the 4d and 5d transition metals has been studied